WATERS Network Cyberinfrastructure Plan

The WATERS Network Project Office Cyberinfrastructure Committee

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Abstract

This report focuses on the evaluation of cyberinfrastructure (CI) needs for the **WAT**er and **E**nvironmental **R**esearch **S**ystems **(WATERS)** community and the development of a conceptual model of CI for environmental engineering and hydrology research and education. The WATERS Network is envisioned as an observatory system that will engender a community-wide shift towards systems-level, CI-enabled research, global research coordination, and bi-directional integration of experiment and simulation.

Implementation of the goals of the WATERS Network requires CI to enable communication, storage and analysis of the data that will be acquired from deployed sensing systems and related research projects. The WATERS Network shares with other observatory initiatives the critical need to organize existing data and to capture new data as they become available. Further, data stored for the network must be organized so that it can be effectively retrieved and analyzed by researchers, educators, students, and policy makers with varying levels of computational knowledge and familiarity with different computer systems and software. Also with other observatory plans, the WATERS community shares a vision that CI will enable transformational research, with a shift toward systems-level, CI-enabled research, global research coordination, and bi-directional integration of experimentation and simulation. WATERS CI is in the design stage and will incorporate features to connect data from WATERS activities with those from other NSF environmental observatories and from monitoring programs supported by federal agencies.

The WATERS community expressed consensus through a survey and reports from SEED grants that CI must be (1) usable and well supported, (2) useful for problems of interest to the community, and (3) adaptable to the changing needs of the research and educational endeavors and to the developing technological sophistication of the users. Further, the WATERS community must design and develop CI to assemble the resources required to address the issues driving the science and engineering plan yet recognize that the WATERS Network will not be the sole sensor network or observatory framework being created. For it to be successful and embraced by the community, CI must enable academic and professional segments of the WATERS community to evolve through a transformational process in which modeling and visualization are ubiquitous and easily accessed. This evolutionary process will rely heavily on human implementation and interaction with the following primary components: **high performance computational infrastructure; robust data management for storage, access and sharing; data fusion and mining; model development and sharing for knowledge generation; visualization for value enhancement; and communication, and collaboration and knowledge networking.**

Cyberinfrastructure that supports the WATERS vision will enable environmental engineering and science education and research in the coming decades. This CI must be carefully planned, implemented, and maintained in collaboration with other observatory efforts. First, it must be responsive to the needs of the community, as articulated in this report. Second, it must be extensible and adaptable to the needs that arise as the WATERS Network moves from vision to implementation. And, finally, it must drive research and education activities to new horizons, enabling critical new questions about our environment to be asked and answered.

EXECUTIVE SUMMARY

This report focuses on the evaluation of CI needs for the **WAT**er and **E**nvironmental **R**esearch **S**ystems **(WATERS)** community and the development of a conceptual model of cyberinfrastructure (CI) for environmental engineering and hydrology research and education. The WATERS network is envisioned as an observatory system that will engender a community-wide shift towards systems-level, CI-enabled research, global research coordination, and bi-directional integration of experiment and simulation. .

Implementation of the goals of the WATERS Network requires CI to enable communication, storage and analysis of the knowledge that will be acquired from the deployed sensing systems and other available data sources, as well as modeling, collaboration, and knowledge networking support. Without carefully planned, implemented, and maintained CI the research endeavors planned to address the urgent environmental questions of humanity will fail. This demand for high level CI is shared with many scientific disciplines moving into the 21st century. Throughout our scientific communities, the questions that loom large demand new technology and scientists and engineers trained to take full advantage of this technology.

The WATERS Network shares with other observatory initiatives the critical need to organize existing data and to capture new data as they become available. Further, data stored for the network must be organized so that it can be effectively retrieved and analyzed by researchers, educators, students, and policy makers with varying levels of computational knowledge and familiarity with different computer systems and software. These analyses will be performed by teams of collaborators from different disciplines who are located across the country. In addition, WATERS CI will be capable of assimilating and integrating data collected from other NSF environmental observatories and from federal programs engaged in water resources assessment. This CI approach is an important component of the WATERS CI design and envisioned utility as an integrative system for WATERS research and the related research and monitoring activities of the broader earth system research community. The WATERS Network shares a vision with other observatory programs that CI will enable transformational research, with a shift toward systems-level, CI-enabled research, global research coordination, and bi-directional integration of experimentation and simulation. The tasks undertaken in this report are the evaluation of CI needs for the WATERS community and the development of a conceptual model of CI for environmental engineering and hydrology research and education. The first task was addressed by (1) considering similar programs focused on other domains, (2) fielding a survey to the WATERS research community, (3) evaluating existing activities within the community, and (4) coordinating with other WATERS committees on organizational issues and CI needs for education and social science.

Community Needs Assessment

CI development and deployment within systems designed for observations in the natural environment is not unique to the WATERS Network. A number of initiatives have been launched by NSF and other agencies in the past decade focused on distributed data collection, collaboration and management, and operation of a network of observing stations or interacting scientific activities across time and space. Some of these activities have been explicitly organized as observatories (e.g., NEON, NVO, GEON) but others described, for example, as

Grids or Collaboratories, have also developed core technologies critical to observatories, including high performance computing (e.g., TeraGrid, OptiPuter), date federation (e.g., Earth Systems Grid, BIRN), or informatics (e.g., SEEK, CMCS). Projects in other countries and general open source and commercial software are additional sources of capabilities. For all but the largest of these projects, funding is provided for an average of only 3-5 years without a clear mechanism for operation beyond the project end or for transferring the relatively mature products of these efforts to new communities before work on a next-generation capability begins. For example, many of the current projects will end before the start of the WATERS Network. It is critical that lessons-learned be gathered and coordination before these projects close out so that the network can leverage relevant developments and resources of these other activities.

CI-enabled research and education directed at grand challenges in environmental research underpins the WATERS Network project. However, the research and education community may not be familiar with the opportunities and challenges of CI in this domain. The WATERS CI committee undertook a survey of researchers who would be potential users of this CI. Association of Environmental Engineering and Science Professors (AEESP) and CUAHSI members were surveyed. Success of the WATERS initiative depends upon understanding the insights provided by this survey. Specific results indicate:

- Data access and search of multiple data sources are perceived as the key contributions of CI.
- Researchers are willing to share data, but have concerns that they will not receive sufficient credit from others who use the data.
- Nonstandard and inconsistent data formats are the key barriers to data sharing.
- It is important for users to be able to identify data sources according to specific measures and variables of interest.
- It is important for users to be able to understand the accuracy of data in order to gauge confidence in data sources.
- Hydrologists and environmental engineers have different needs when using data.
- Excel is the most popular application for data analysis, followed by ArcGIS and MATLAB.
- Ease of use and utility are the drivers for technology adoption.
- Access to expertise, particularly complementary expertise, motivates the search for collaborators.

These specific results are summarized in three key points. First, CI must be usable, saving more time than it consumes. The design of CI must be clear and appealing, something with which users can readily identify. Sufficient support for training, maintenance and trouble-shooting must be available. Second, CI must be useful. The survey respondents identified some low level

access problems that could be easily solved and result in significant improvement for users. Finally, CI must be adaptable to the changing needs of the research and educational endeavors and to the developing technological sophistication of the users. What is needed and works well today may not meet the needs of tomorrow. For example, as sensors begin to produce larger quantities of data, analyzing that data only with Excel will likely not be feasible. Nonetheless, in designing CI to address such future needs, the qualities of existing tools that led to their current popularity should be evaluated and considered. For example, Excel could be popular because it is general enough to use across projects and thus justifies the investment in training on it, it enables custom exploration without the complexity of full computer programming, and it is popular enough to serve as an exchange mechanism with others. CI must enable users to explore beyond the limits of whatever observatory structure is implemented, and support end-user exploration and evolution.

CLEANER, the environmental engineering community's predecessor to the WATERS network was the focus of targeted seed grants in 2002. Many of these projects were focused on specific field sites and considered aspects of cyberinfrastructure necessary for success at a field site in a large collaborative network. Review of these projects focused specifically on CI aspects of the proposed and executed work and clarified several critical features of CI in the environmental engineering and science domain:

- WATERS community members hold different definitions of CI; the community needs a clear vision of CI in order to move forward.
- Data collection, management and access drive CI needs during the early phases of planning and implementation. This aspect of CI represents a universally-acknowledged need within the environmental engineering and science research and education community.
- Poor CI usability restricted research agendas and slowed progress. Current-off-the-shelf CI and sensor technologies lack ease of use.
- CI represents an important support need for the WATERS network, but application of CI in the environmental domain also presents challenging research questions for collaborators in computer science. Understanding the IT domain and framing environmental CI needs within established areas of IT research will facilitate greater collaboration among primary researchers in both domains.
- Students require cross-disciplinary training to enable them to engage in CI development or application of CI to environmental research.

Each WATERS committee addressed one aspect of the WATERS vision: science plan, social science plan, education plan, organization plan, sensor needs, and cyberinfrastructure. The CI needs for these different aspects of the WATERS vision are different but related. The education committee focused on two aspects: training needs of the community and using CI with pre-college and university students. The design of tools for these two educational objectives should be developed in tandem with the larger system of tools designed to meet research objectives. The design of CI must consider how resources effectively translate into materials that are practical and meaningful to teachers and students. The social science committee focused on collection of human societal data that was complementary to the hydrological and water quality data that is the focus on much of the science plan; incorporation of CI needs for this community involves

adding to existing data networks with slight modifications. Social scientists more extensively use survey data and access different resources (e.g., census), and thus, CI must accommodate this type of information and these types of cross-disciplinary queries. The organizational committee focused on location and operation of CI; the degree of centrality in the human and computing resources for CI must be decided prior to implementation.

Conceptual Model of CI in WATERS

The second task undertaken by the WATERS CI committee was to propose a conceptual model for CI within this community. There is significant challenge to a conceptual model. Specific needs identified through the survey must be met, while avoiding an over-engineered solution. That is, CI must be designed for both scientific and technological progress and focus on scalability and evolvability rather than aiming for an all-encompassing scale and scope. The WATERS community must design and develop CI to assemble the resources required to address the issues driving the science and engineering plan yet recognize that the WATERS Network will not be the sole sensor network or observatory framework being created. Researchers will need a well managed set of core capabilities yet will also desire the ability to include new data sources and tools (from local projects or other national/international projects).

The prerequisite for all other functionality of the WATERS observatory CI is the ability to capture data from the proposed sensor network, calibrate and apply quality control procedures to it, and provide ongoing access to it for the broad WATERS community. This leads to requirements in terms of networking, computational power, and data storage capabilities as well as software to support the overall data flow, to manage and monitor the system, and to provide effective data discovery and access. It also speaks to the need to be able to link to relevant data sets produced by other NSF environmental observatories and to the hydrological research and monitoring programs of several federal agencies. However, this set of requirements is not sufficient on its own to meet the larger goals of the observatory. To enable the development of new models of environmental processes and human impacts, the CI must support the incorporation of data directly into the full range of relevant models, the calibration of models, and the comparison of model predictions with each other and further observations. To serve as a community focus, the CI must support scientific discourse and support the large-scale dissemination of new ideas and resources. To serve as a model for future sensor networks and long-term, large-scale scientific data curation efforts, the CI must demonstrate best-practice capabilities that are independent of the specifics of environmental research and that can be cost-effectively adapted to changes in technologies and in scientific goals over decades.

The WATERS CI must be characterized both in terms of component functionality in areas such as data management and collaboration capabilities as well as in terms of overall, cross-cutting system qualities. Several such aspects are critical for environmental observatory CI: interoperability, quality control, scalability, balance, and extensibility.

- The WATERS Network will incorporate resources managed by other projects and agencies, co-exist with a range of other environmental observatories, and live in the context of the broad development of CI and CI-enabled science and engineering. Interoperability and re-use across this ecosystem is clearly required to enable the

WATERS science plan as well as to build and operate the network in a cost-effective manner.

- Successful implementation of cyberinfrastructure (CI) for the WATERS Network will require rigorous validation and testing of applications and services.
- When building a permanent facility, a critical element of planning is to design its infrastructure to meet not only current needs, but the anticipated needs over the full lifetime of the project. Traditionally, in sizing a facility building for example, this involves initial over-provisioning. With cyberinfrastructure for observatories, this can be an extremely costly approach due to the combination of exponentially decreasing costs and exponentially increasing demands.
- The WATERS community uses a wide variety of models and visualization tools in the academic pursuits of teaching, research, and outreach; and in the provision of professional services in private industry and public service. In order for CI to be embraced by the environmental engineering and science community, it must provide an immediate perceived benefit, such as time reduction and streamlining of modeling and visualization tasks that lead to project goals.
- Similarly, the WATERS community will continue to evolve their research and require additional capabilities from the observatory over time. The CI will need to enable innovation and extension of the system in parallel with production use.

While a detailed technical architecture for the WATERS CI cannot be specified without further investigation and additional detail on the specifics of the science plan and other aspects of the WATERS conceptual design, the general scope and cross-cutting concerns described above do constrain the design space sufficiently to identify an appropriate set of design principles and an overall conceptual architecture and component-level decomposition.

At this level of specification, the WATERS CI should be an anticipated-best-practice design for a scalable, distributed, multi-objective CI for long-term scientific knowledge creation and dissemination as part of a larger knowledge ecosystem. Framed this way, it is clear that the WATERS Network can draw from CI design efforts from many disciplines, particularly those that address issues of long-term persistence, evolvability, and interoperability, and conversely, that the discussion and design information gathered here should be relevant for other environmental observatories and next-generation community cyberenvironments in general.

At the most fundamental level, the WATERS CI must be architected for change: change in scale, change in component location, change in data types and community processes, change in scientific understanding, change in technologies, change in the community, and change in the overall scientific ecosystem. For example, increases in computing power and data storage are expected, as are decreases in the cost of sensors and networking. Many technologies we now take for granted (e.g., web browsers) were developed, deployed, and became readily available in less time than will exist between release of this planning document and the expected implementation of the WATERS Network. The most appropriate design response, again at a fundamental level, is to incorporate capabilities for managing change in the system and to decouple the most rapidly changing aspects from core capabilities and from each other. For example, if resource location can change, the system should use global identifiers that are independent of location and provide a resolution mechanism to provide a current location and

access method(s) for a resource given its identifier – the basic idea of address books and phone directories.

For the WATERS CI to be successful and for it to be embraced by the community, the CI must enable academic and professional segments of the WATERS community to evolve through a transformational process in which data, modeling, and visualization become ubiquitous and easily shared. This evolutionary process will rely heavily on human implementation and interaction with the following primary components:

- High performance computing-based infrastructure;
- Intensive data management for storage, access and sharing;
- Data fusion and mining;
- Model development and sharing for knowledge generation;
- Visualization for value enhancement and communication, and
- Collaboration and knowledge networking.

This report provides additional detail on the functionality required in these component areas but stops short of identifying specific levels of performance. For example, the draft WATERS science plan indicates significant interest in real-time analysis, modeling, and adaptive feedback of the observational system to better understand and ultimately manage event-based environmental phenomena. This leads to requirements for on-demand processing triggered by the arrival of new data, but the overall computational requirements for real-time streaming and processing (QA/QC, creating derivative data products) of WATERS observations are uncertain. While a small network of sensors sampling infrequently could require relatively modest memory and computation cycles, many aspects of the planned program could require significant traditional or on-demand high performance computing resources. The resources needed to respond to requests for historical data, to create custom data products on demand, to run high-fidelity models for comparison with observation, or to directly apply modeling to optimize sensor positioning and overall experimental design or perform research (e.g., for predicting the performance of potential management practices) could require large numbers of processors and would grow as the overall knowledge base and user base expand. Depending on the specific use case, the computations may involve traditional batch processing on shared resources such as the TeraGrid, or they may require the coupled on-demand availability of computation, network bandwidth, and storage resources for event-based research. Thus, at this stage, it is difficult to predict the exact scope of HPC requirements or the expected growth rate as WATERS-derived science expands, though it is clear that both types of computation should be part of the overall plan. Similar analyses has been done across the CI to detail the mapping from requirements to specific types of functionality in each component area, though again, moving from this level of detail to a technical system specification will require additional information.

A cyberinfrastructure developed based on the conceptual architecture presented in this report will be well suited to supporting the WATERS vision and enabling environmental engineering and

environmental science education and research in the coming decades. This CI architecture identifies the technical capabilities that will be necessary for managing the observatory network and data, for supporting data dissemination and analysis, and for enabling community coordination to address the WATERS research and education mission. It also incorporates design concepts that will support the creation of a well-engineered, carefully planned, robustly implemented, and cost-effectively maintained infrastructure that can leverage other observatory efforts. Most importantly though, the architecture will be capable of meeting the needs of the community and can be built incrementally in a manner responsive to the priorities of the community. Further, it is extensible and adaptable and will be able to meet the needs that arise as the WATERS project moves from vision to implementation. And, finally, as with the overall concept of the WATERS Network, the proposed CI represents a dramatic leap in capabilities that will inspire the community to drive research and education activities to new horizons and enable new questions to be asked and answered.

WATERS CYBERINFRASTRUCTURE PLAN[2]

I. Introduction

The **WAT**er and **E**nvironmental **R**esearch **S**ystems **(WATERS) Network** requires cyberinfrastructure (CI) to advance research and educational objectives driven by the grand challenges outlined in the WATERS Science Plan. As part of the **WATERS Network** planning process, the Cyberinfrastructure (CI) Committee is tasked with gathering the CI needs for the research and education community, outlining a conceptual model of CI development for this community, and where possible, defining a structure for the CI implementation within the WATERS Network. The committee responded to the first part of this task by (1) considering similar programs focused on other domains, (2) fielding a survey to the WATERS (hydrology and environmental engineering and science) research community, (3) evaluating existing activities within the community, and (4) coordinating with other WATERS committees regarding their anticipated CI needs. Further, the NRC Review of the WATERS planning effort was released during the committee's work, and the concerns of this group were also reviewed.

The second task of defining a conceptual model for CI development for this community took as its starting point the results of the needs assessment. Three overarching themes form the basis of the conceptual model development:

- CI should be enabling and not stifling to research and education objectives. Evolution of CI and applications is expected, and this expectation should be incorporated within the conceptual model.

- CI must be innovative and scaleable. Staged implementation and a distributed development model will produce the best results.

- CI investment must match planned goals. Limited resources will require prioritization to enable research goals while also fostering inclusivity and community buy-in.

The conceptual model proposed in this report builds on these three "take-home" messages. While there was a strong desire to provide a specific structure for CI implementation for the network, the committee was unanimous in its belief that this structure must evolve through direct integration of the science, social science, education, modeling and sensors plans for the WATERS initiative. Further, this structure must be developed in collaboration with other communities active in CI for the environment.

This report focuses on cyberinfrastructure. There are two additional and related committees as part of the WATERS planning effort: the modeling committee and the sensors committee. Thus,

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[2] The Cyberinfrastructure Committee was co-chaired by Tom Finholt (University of Michigan) and Jeanne VanBriesen (Carnegie Mellon University). Members included: Russ Hobby (Internet2), Monica Mazurek (Rutgers University), Barbara Minsker (University of Illinois at Urbana-Champaign), James Myers (National Center for Supercomputing Applications), Michael Piasecki (Drexel University), Wayne Woldt (University of Nebraska), Ilya Zaslavsky (San Diego Super Computer Center)

modeling and sensors are considered related to but not components of CI for the purposed of this report, and the reader will be referred to the other committee planning documents for details of these critical components.

II. Background

Cyberinfrastructure is a term coined by the Blue Ribbon Advisory Panel that advised the National Science Foundation on the implications of information technology on science and engineering research. (Atkins 2003). The term is now used widely to refer to a range of computing resources.

CI has been defined as computers, networks, archives, grid services, collaboration services, information technology services, data management, data mining and visualization services. (Atkins, 2003). Figure 1 (reproduced from the Atkins report) incorporates these components into layers, with the community-specific collaboratory at the top, supported first by custom, discipline- and project-specific applications.[3] Under these critical components are the information technology services that support them: high performance computing, knowledge management, observation and measurement, interface and visualization, and collaboration. Even more fundamental are the networking, operating systems and middleware and the computation, storage and communication protocols that form the base of the CI. The component-centric view from this figure and the Atkins report more broadly have provided the reigning definition for CI within scientific communities for several years.

Over the past year, the NSF Cyberinfrastructure Council has been working to transform the grand vision of the Atkins report into a strategic plan for cyberinfrastructure components. This report reemphasizes the definition of cyberinfrastructure in terms of its effect: “the comprehensive infrastructure needed to capitalize on dramatic advances in information technology.” This view is in keeping with Ian Foster’s suggestion that CI is fundamentally service-oriented, automating time-consuming activities and allowing scientists to spend more time on interpretation and development of insights into their data. (Foster, 2005)

A definition encompassing both the critical components and the outcomes would be “the set of reliable, well specified and interoperable connections of electronic hardware and software that allows people to discover, learn, teach, collaborate, disseminate, access, and preserve knowledge in their domain.” (NCAR 2003)

These definitions should not be seen as describing different concepts. Instead, they describe the same concept from the perspective of cyberinfrastructure developers and from that of scientific users. The mixed definitions describe the cyberinfrastructure’s value to the domain-based research audiences and then map that to implementable components and functionality, and thus serve as a minimalistic conceptual architecture. This report is intended to serve the same purpose with a more comprehensive definition of cyberinfrastructure in the context of the WATERS Network that clearly links what cyberinfrastructure is and what it can do.

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[3] The Atkins report did not consider these two top layers as cyberinfrastructure; however, they have increasingly been included in reports from research-specific communities. The base technology (computation, storage, communication) was also not considered CI in this report.

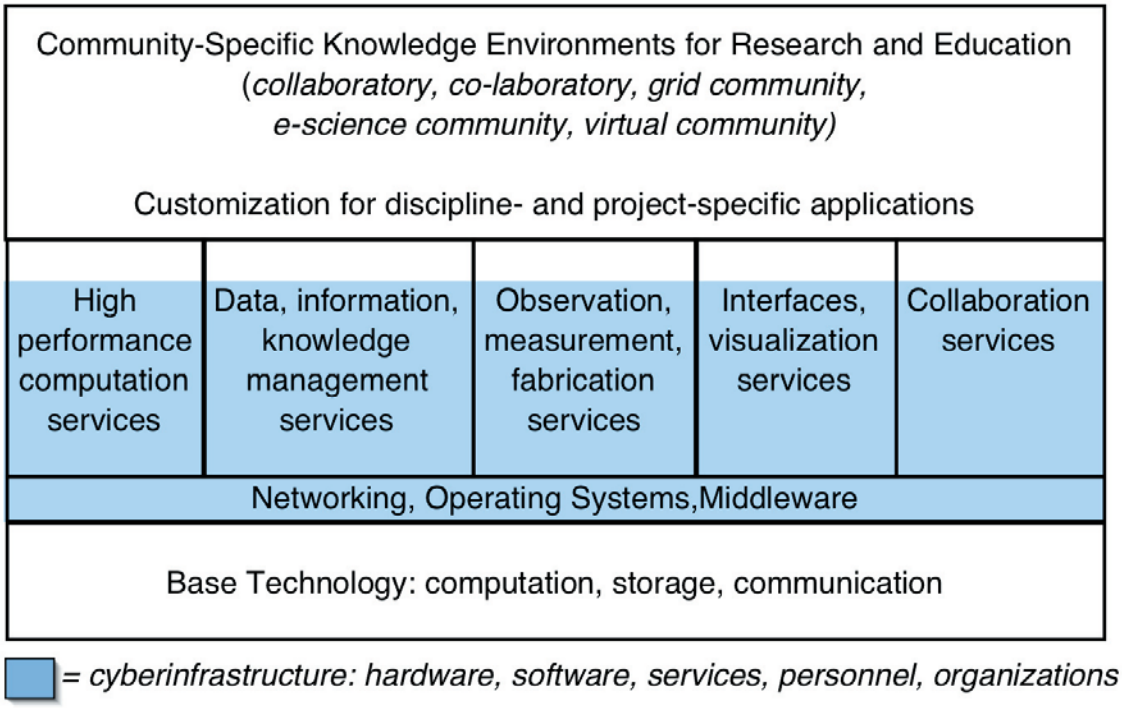

Figure 1. Conception of cyberinfrastructure from Atkins et al. (2003)

III. CI Needs Assessment

To consider the needs of the WATERS community related to cyberinfrastructure, we began by considering recent CI initiatives within NSF and the programs and observatory networks that are being developed concurrent with WATERS efforts. Following that, we considered efforts that have been seeded within the WATERS community (focusing mainly on the CLEANER planning grants as the CUAHSI planning efforts have been documented elsewhere) (Maidment, 2005). Finally, we summarize the results of a survey that was fielded to the WATERS research community.

A. NSF's CI Plans and other Observatory Initiatives

NSF has had a leading role in the development of cyberinfrastructure across scientific disciplines. The history of NSF's investments in CI is detailed in Appendix B of the NSF Vision (NSF 2006). Briefly, NSF's investment began in the 1960s and 1970s and was focused on campus-based computing facilities. In 1985, NSF launched its Supercomputer Centers program (leading to five academic-based centers). NSFNET was developed in the 1980s. In the 1990s, vBNS and the Middleware Initiative were launched. In the 1990s the PACI (Partnership for Advanced Computation Infrastructure) Program was launched and the high performance computing (HPC) was organized to address "grand challenges." Later in the 1990s, several significant CI projects were launched including the Network for Earthquake Engineering Simulation (NEES), the Grid Physics Network (GriPhyN), the International Virtual Data Grid Laboratory (iVDGL) and the Geosciences Network (GEON). Early in the 2000s, the Extensible Terascale Facility (TeraGrid) was initiated and two Supercomputer Centers (San Diego Supercomputing Center and the National Center for Supercomputing Applications) were tasked with better defining CI needs through close engagements with national science and engineering

communities. Most recently, NSF has been seeding groups like CUAHSI, CLEANER, NEON, OOI, and now WATERS to evaluate the role of CI in their communities and to develop large scale observatory systems for the natural environment. A list of reports from these activities is provided in the Appendix A. NSF has also funded a wide range of CI-related developments in many disciplines through programs such as the Knowledge and Distributed Intelligence (KDI) and via domain-focused programs in areas such as Cyberchemistry. Complementing the NSF portfolio have been investments made through the Department of Energy, National Institutes of Health, National Aeronautics and Space Administration, and by other countries aimed at developing collaboratories, grids, e-science, and general HPC and network infrastructure. NSF researchers are an active part of this larger community and there is significant sharing of best practices and software components as well as formal standards efforts that bridge these efforts.

While the nascent NSF observatory communities have been meeting and discussing CI needs for several years now, there is a critical need for a team to come together across these observatory programs to plan for the interoperability and cross-utilization of resources among the programs. This was highlighted by the recent National Research Council review of the plans for CLEANER (now WATERS) (NRC, 2006). An overarching parent organization was recommended to coordinate and integrate all environmental observatories and their CI needs. The organization would be an "Environmental Observatory Network" (EON) that would be charged with the development and integration of basic observatory components such as CI as well as educational, outreach and other shared activities. One key function EON might perform would be to facilitate connectivity between the environmental observatories and to reduce redundancy.

Based on this recommendation, the CI committee for WATERS reviewed existing research observatory networks and other relevant scientific CI projects. This includes projects explicitly organized as observatories (e.g., NEON, NVO, GEON) as well as those developing and deploying core technologies that are critical to observatories, such as high-performance computing (e.g., TeraGrid, Globus, OptIPuter), sensor network management (e.g., ROADnet, LOOKING), data federation (e.g., Earth Systems Grid, NEES, BIRN), or informatics (e.g., SEEK, CMCS).[4] Several things are notable from a management perspective. First, there is significant range in the amount of funding – from a few hundred thousand dollars to more than five million per year with durations from two to ten years. Projects also differ in focus – on domain-independent capabilities versus integration of such capabilities to serve a domain community – and on their place in the technology lifecycle – research versus pilot deployment versus operations. Further, projects span the directorates at NSF; with the exceptions that SBE and CISE don't fund any awards (although it could be argued that the Next Generation Cybertools awards made by SBE in partnership with OCI are comparable to the smallest projects described here).

Such a diverse ecosystem of CI development at multiple levels and for multiple uses clearly provides complementarity, but also increases the need for communication among projects to avoid effort duplication and to improve interoperability of systems. Continuing activity throughout NSF (and DOE, NIH and international partners) indicates broad interest in and

[4] This list is not inclusive. It is presented for illustrative purposes only.

support for CI development, and again highlights the need for communication and coordination to avoid duplication and enhance collaboration. However, due to the relatively short lifetimes of most projects, many of the most relevant current projects will end before the WATERS Network begins operation. It is critical that current projects capture their lessons-learned, open source their software and be encouraged to compete for new funding to maintain and grow the expertise that has been developed. This will benefit all the observatory programs, including the WATERS Network.

Three observatory initiatives are directly related to the objectives of the WATERS Network – the Consortium of Universities for the Advancement of Hydrologic Science (CUAHSI) Hydrologic Observatories, the National Ecological Observatory Network (NEON), and the Ocean Observatory Initiative (OOI). Plans for these observatory initiatives were reviewed in detail, and summaries of these reviews are presented in Appendix B. Related directly to CI, we make the following conclusions from this effort:

- Cyberinfrastructure is proposed for data management, integration of data streams from sensors, and modeling in all the observatory programs.
- There is a consistent need to deal with long-term high frequency data from sensors. The need to conduct episodic event-driven science as well as to study processes that occur over decades is a focus of all the efforts.
- Standardization of data collection and provision; software standards; and access issues across different networks is a priority for all the observatories.
- CI is expensive; all the programs recognize the need to leverage a shared cyberinfrastructure with other observatories.

As the WATERS Network is envisioned as the joining of the CLEANER initiative and the CUAHSI network, this committee closely evaluated the work of the CUAHSI Hydrologic Information Systems (HIS) group. CUAHSI HIS has the following conclusions regarding CI for hydrology that we endorse as being broadly applicable for CI for WATERS:

1) The hydrologic community relies to a large extent on federal data collection systems, including observation networks organized in USGS's NWIS and NAWQA, EPA's STORET, Ameriflux tower network, MODIS and DAYMET datasets, and others. These data are in the public domain, and these repositories are freely accessible via respective web portals. This has at least three consequences for the design of cyberinfrastructure: (1) making access to such repositories simpler and more uniform (based on a community observation data model), would immediately benefit a large group of research hydrologists, as was revealed by a CUAHSI user survey[5], (2) it is important to engage with respective federal agencies, to make access to their repositories robust, uniform and sustained, and (3) emphasis on data ownership and data security is relatively weaker in hydrologic analysis as compared with other research communities. This justifies the focus

[5] This was echoed by the WATERS survey detailed below.

on a common web service interface and a hydrologic data access portal easing access to federal observation network archives, without the necessary user authentication as is customary in other portal environments.

2) The hydrologic community appears to be organized, to a larger extent than other geoscience communities, by "natural" boundaries, specifically by river watersheds and sub-watersheds. This suggests a "natural" network of relatively autonomous hydrologic data nodes that provide access to locally-collected and curated data resources and applications. Therefore, development, deployment and technology support of such nodes is an important component of creating a networked environment for hydrologic data sharing.
3) From the data perspective there are sub-groups in the community focused on analyzing point time series (and incidentally relying largely on Windows platform) and focused on analyzing remote sensing data and time series (and using Linux/Unix platforms to a larger extent than the first group). Supporting different groups of researchers requires that HIS relies on cross-platform data management services, and portals that can be deployed in both environments.
4) Given the focus on water resources, CUAHSI communicates mostly with public sector entities such as local water authorities and related small engineering firms. This creates a lot of opportunities for partnerships at the local level, and underscores the need for a data access infrastructure supporting such partnerships.
5) As revealed by CUAHSI user survey, the community relies on several common COTS (commercial off-the-shelf) software packages, most importantly Excel, ArcGIS and Matlab.[6] Enabling access to time series repositories from these desktop applications, as well as from such popular coding environments as Fortran and VisualBasic, is an important consideration for HIS architecture.
6) Hydrology is an integrative science, with hydrologic models relying on data inputs from several neighboring disciplines (climate and ocean observations, soils, geomorphology and geology, social and demographic datasets, etc.). Consequently, the HIS infrastructure shall support interoperation with data and processing services being developed in other earth science disciplines, and ideally develop similar formats for handling spatio-temporal information.
7) Different hydrologic analyses may require different representations of space and time. For example, hydrologic time series services may need to expose observations in both local and UTC time: the former is common at large scale watershed-level studies while the latter may be needed for compatibility with atmospheric data services. The same applies to handling spatial locations of hydrologic observations, where multiple types of offsets from hydrologic landmarks are commonly recorded.
8) Large numbers of observation variables, on water quality in particular, available in federal repositories (there are nearly 10,000 variables measured within USGS NWIS only) and often inconsistent semantics of variable and measurement unit descriptions across observation networks make the development of observation data catalogs and knowledge bases, and semantic mediation techniques, indispensable.

[6] Again, see confirmation of this for the whole WATERS community in the survey results presented below.

9) Given the reliance of water researchers and practitioners on several commercial software packages, it is important to engage with research and user services groups of respective software market leaders (e.g. Microsoft, ESRI, Kisters) and standards groups (e.g. W3C, Open Geospatial Consortium) to ensure compatibility and standards-compliance, and reduce duplication of efforts with commercial concerns. The CI is envisioned as an evolving system that can accommodate a mixture of software licensing models. Leveraging resources of software leaders, and their market reach, would ensure that the developed CI is integrated in regular hydrologic analysis and environmental engineering workflows and will be sustained beyond the life of the project.
10) In addition, it is important to support software development community and a collaborative environment where open source hydrologic software projects are encouraged and sustained, and interested developers have a chance to contribute to the CI – via sharing code, software development templates and guidelines, etc.

B. Community CI Needs Assessment

Although the potential uses and benefits of CI are in the spotlight at NSF, the actual CI requirements of practicing environmental scientists and hydrologists are less well articulated. In part, this is due to limited communication between practicing researchers with needs and CI developers with technologies to help build solutions. To bridge this communications gap, an ongoing dialog to analyze requirements and interests in terms of economically and technically feasible infrastructure will be required to enable new science and engineering through the adoption of these technologies by domain scientists and engineers.

To this end, we conducted a survey of researchers who would be potential end users of the WATERS CI. The survey was intended to capture information about three aspects of CI and its relationship to the environmental engineering and hydrology community: data needs, technology use and adoption, and preferences for collaboration. Because the most immediate gains from the use of CI will likely be in the acquisition, manipulation, and storage of large datasets, understanding users' data needs and priorities are important. Similarly, identifying expectations and norms of technology use will allow developers to deploy CI in a way that will attract an initial set of dedicated users. Finally, some of the currently available tools and capabilities of CI are those that allow people to share documents and files, interact online, and search for colleagues and resources; therefore, knowing users' preferences for collaboration will make sure that the WATERS CI focuses effort on providing the most sensible tools to support their work. In short, the survey sought to identify how cyberinfrastructure could enable environmental engineers and hydrologic scientists to conduct research and collaborate in new ways. A limitation of the survey was that it was designed to capture data about how the community could move incrementally from current practice to use more CI. Qualitative changes in practice (such as the ability to run another investigator's program seamlessly on your own data from your own desk) are difficult to articulate in a survey of this type. Articulating the possibilities that will cause broad, large-scale shifts in the approach of domain researchers to CI is one of the goals of this planning document.

Details of the survey, its results, conclusions, and a complete list of its questions are provided in Appendix C. The major results are summarized here.

CI must be usable: Users will have little patience with an application that consumes more time than it saves. Too often software developers overlook the importance of human-computer interaction (or HCI), assuming that the perspective of the user is the same as their own (Cooper, 1999). The design of the CI must be clear and appealing, something with which users can readily identify. Behind the system must also be sufficient support for troubleshooting, training, and maintenance.

CI must be useful: Developers can achieve the fastest gains by focusing on small wins—basic resources that are easy for newcomers to recognize and embrace. Resources apparently do not need to be high-end or unusual to offer a distinct benefit for users. Simple access to data and storage will accommodate many scientists' needs for both research and collaboration.

CI must be adaptable: Although people think they know what they want now, there is the possibility that new capabilities and technologies will surface over time. Even with current predictions of software use, requirements for ten years in the future cannot be predetermined today. Consequently, the underlying architecture of the system must not be pinned to proprietary technologies that are too rigid to adapt to future contingencies. As one person aptly put it: "I am concerned that infrastructure for its own sake does not move science and engineering forward. It's not clear that the community has decided what questions are to be answered. Do that first, and then decide what infrastructure is necessary."

Finally, in addition to these conclusions related to general CI systems, the respondents were consistent in their data focus. **Finding, accessing, and validating data sources were significant limitations to further use of CI.** A clear message from the community was that improved data archiving and access is a critical first step in developing a CI for the WATERS Network.

C. CLEANER Planning Grants

In 2002, NSF funded twelve seed grants to enable researchers to test ideas related to the creation of the CLEANER network, the environmental engineering community's predecessor to the WATERS Network. These grants considered environmental engineering and/or cyberinfrastructure research questions. Many were focused on specific field sites and considered aspects of cyberinfrastructure necessary for success as a field site in a large scale collaborative network. The committee reviewed the original proposals, the progress reports, any written products, and participated in discussions with the Principal Investigators at presentations of research progress made in early 2006. Specific, project-based summaries are provided in Appendix D. These provide details of the CI proposed, implemented and lessons learned during these planning grants.

Review of these planning grants highlights several critical features of cyberinfrastructure in the environmental engineering and science domain.

- The WATERS community continues to have many different definitions of CI. Some are very general, including any IT. Others are very specific to sensor hardware and software issues. Others are strongly focused on data handling and management. It is critical that a definition of CI, even if it is very broad and sub-divided, be put forth by the WATERS Network so that education of the community can be consistent. The community needs to

see the value of CI and to do that a consistent vision of CI must be put forth by the WATERS design committees. This is the objective of the present document.

- **Data collection, management and access are a primary component of CI in all of the planning grants**. This aspect of CI represents a universally-acknowledged need within the community. The issue of multiple data formats and standard database structures, thus, must be addressed by the WATERS Network.

- The link between cyberinfrastructure and sensor networks is strong and necessary as many of the environmental field facility (EFF) sensor networks will generate significant quantities of data that must be managed by the CI. Discerning where planning for the sensors and their control ends and where planning for the CI begins will be difficult. **The CI committee strongly recommends that the WATERS planning consider the sensor communications and the CI data structures as a single, unified development need.**

- Engagement of information technology *researchers* with critical problems in developing CI for the environmental domain requires attention to the novel aspects of environmental CI such that the IT researchers are presented with interesting research questions as well. Development of CI aspects that are not novel and just need to be adapted to the environmental domain should be left to IT professionals not researchers. The development of CI for the WATERS Network will require input and collaboration from both IT professionals and IT researchers.

- Environmental engineering graduate students require cross-disciplinary training if they are to engage in CI development or application to environmental research. Many environmental engineering professionals likewise must allocate time and effort to learning the language of IT collaborators. Additional education at the disciplinary boundaries between environmental engineering, hydrology and information technology is needed. However, it is also critical that usability of the developed CI be high so that obstacles to use and further development by environmental engineers and hydrologists do not discourage research.

- Usability of sensors and CI was highlighted as a critical issue that must be overcome to ensure that as many domain scientists as possible can become engaged with the WATERS Network. Current off-the-shelf products still lack ease of use. A focus on use case studies and early requirements testing can contribute to development of usable CI.Collaboratory efforts were proposed in many projects; however, most included only web site development. In general, challenges in other CI areas limited effort on collaboratory development. Extensive collaboratory development requires CI expertise as well as domain interest.

D. Broader Community Needs Assessed via other WATERS committees

In addition to the survey (section B above) and the evaluation of WATERS planning grants (section C above) the community needs were also assessed through discussions with the WATERS committees tasked with Education, Social Science, and Organization.

Critical Requirements for Users in Educational and Outreach Settings

Cyberinfrastructure development and use must address educational issues in two major ways. First, for diverse communities to effectively participate in the WATERS network, some degree of training in the use of the CI will be necessary. Thus, while creating a system that is as user-friendly and accessible as possible, developers of the network should likewise consider how the CI will be supported by educational materials that are accessible to end users. Second, the CI needs to effectively identify and meet the needs of educators who will use the WATERS Network with pre-college and university students. Teachers and instructors may also find that the WATERS CI provides a valuable resource for their own professional development. A key opportunity is for educators to work together by sharing knowledge, resources, and methods with each other. Such collaboration can make the adoption and use of these materials more efficient.

To accomplish these goals, several processes (and adequate supporting funds) need to be in place. The design of tools and resources for educators must be developed in tandem with the larger system targeted at researchers. Although the participation of multiple stakeholders adds to the complexity of the design process, it also ensures that the system will be thoughtfully developed with the needs of multiple users in mind, rather than adding educational functionality as an afterthought. As part of this process, developers will need to conduct a requirements analysis that identifies who will use the system for educational purposes, as well as how, when, and where it will be used. In addition, the CI may need to engage people who can help specialists in one area of research to understand how to make their data or materials more accessible to non-expert audiences such as teachers or high-school students. Part of this translation process may be helped by considering the standards for K-12 math and science education and identifying how the use of the WATERS network resources will help students achieve those learning goals. An example of an online resource that takes this approach is http://www.teachengineering.com/. In sum, developing a system that will be usable and appealing to educators will require an iterative, responsive design process that considers how resources can be effectively translated into materials that are practical and meaningful for teachers and students. Note that requirements for education should also consider what might be helpful for outreach activities, such as the potential for incorporating data collection by individuals ("citizen science") or making the observatory findings accessible to policy makers.

Critical Requirements for Users in Social Science

With forethought, cyberinfrastructure will be able to support the activities of social scientists who participate in the WATERS Network. In some cases, the data collection needs of social scientists may be closely related to the types of data being collected by the physical and natural scientists who participate in the network. In other cases, complementary data may need to overlay existing digitized information; for example census data to overlap with water supply data. Social scientists may also need to connect with other types of scientists whose research is related to their own. In all these circumstances, for CI to be effective, social scientists and economists must be involved early in the development of sensors, data systems, and collaboration tools. For example, it is possible that sensors may be readily adapted to gather supplementary data for social scientists to use, without affecting the other data collection activities of a sensor. Social scientists also rely more heavily on the use of surveys for data collection, so data management systems should be compatible with survey data. In addition, data resources should allow interaction between geographic information systems and census data,

Bureau of Labor statistics, and similar data sources that are not typically used by environmental engineers or hydrologists. In sum, the hurdle of accommodating the needs of social scientists and economics should not be great if the similarities and differences of their needs and interests, as compared to those of environmental engineers and hydrologists, are identified early in the process and incorporated into the design of CI systems.

Organization and Administration of Cyberinfrastructure

As cyberinfrastructure is developed to support the WATERS Network, one concern is how this CI will fit within the operational and administrative structure of the network. Indeed, part of the organizational structure will involve people whose primary task is to develop, maintain, and service the CI and other technologies being used across the country. Because this infrastructure will serve people with diverse needs and interests, but those people will require a certain degree of coordination across sites, a pertinent question is whether CI personnel and functions should be centralized or decentralized. This issue is quite fundamental to information technology (IT) deployment, and many organizational objectives drive the debate as to whether computing resources should be centralized or not (King, 1983). For example, centralization is wiser when an organization needs to contain costs, standardize services, ensure coordination, and exert control over IT professionals. In contrast, decentralization is important if individual organizational units need autonomy, have special needs, or require quick responses. Historically, because IT divisions often have expertise that is not duplicated among other divisions in the organization, thereby creating a relative monopoly in the provision of services, IT divisions may tend to fulfill the immediate needs of their own work goals at the expense of being responsive to the demands of the rest of the organization (Danziger, 1979). Centralization may aggravate this tendency. A report by McClure, Smith, and Sitko (1997) suggests that centralized IT structures are not typically adequate to meet the needs of a diverse, distributed audience, and the most effective approach is to provide local support services with departmentally-situated IT consultants. Compatibility and interoperability is secured through a standard baseline environment, supported by a central group that also provides training and resources to the distributed consultants. This arrangement allows better tailoring of services, support, and expenditures because local consultants understand the local unit's needs, personalities, and culture. Critical to such a structure, however, is adequate funding to provide the centralized resources (e.g., training materials for consultants to use, tool development frameworks) that can support the locally situated consultants. Moreover, one has to consider how well such a distributed model applies to a national observatory network. For support of observatory activities (i.e., data collection and archiving), the previous findings on distributed IT would clearly be applicable, but the IT also needs to support the national community who are distributed across all universities. Therefore a mixed model may be needed, including both decentralized staff at the observatories and more centralized staff reporting to the central network office, who would ensure consistency across observatories and address the needs of the broader community. These issues are the focus of the WATERS organization committee and are further discussed in that planning document.

E. NRC Review of CLEANER Activities

The NRC' Water Science and Technology Board convened a committee to review CLEANER and its merged partnership with CUASHI, now collectively defined as WATERS. One design concern relating specifically to CI was "software security failure resulting in data loss or major

delays in research." The committee recommended envisioning possible scenarios which could help with the design of data and software systems that would provide "reliable long-term storage and management of time-series and other data." Overall, software storage and security was judged to be a serious concern not only for CLEANER, but also for other environmental observatories (e.g. NEON, GEON, OOI).

An overarching parent organization was recommended to coordinate and integrate all environmental observatories and their CI needs. The organization would be an "Environmental Observatory Networks"(EON) that would be charged with the development and integration of basic observatory components such as CI as well as educational, outreach and other shared activities. One key function EON might perform would be to facilitate connectivity between the environmental observatories and to reduce redundancy. CI was considered a key component in integrating the observatories and maximizing successful outcomes. The committee strongly recommended CLEANER (and thus the WATERS Network) integrate its efforts with Federal monitoring programs and with the wealth of environmental data collected by water research programs supported by NSF, USGS, USEPA, NOAA, US Fish and Wildlife Service, USDA, DOD, DOE and NASA. The task of integrating the programs and data streams, databases and models with those that will be produced through CLEANER's activities will require a "robust and adaptable cyberinfrastructure." The integration with federal water research programs should be an early program objective along with specific CI development to enable the links.

The committee identified three essential design features of CLEANER's environmental observatories. Two of them are directly relevant to our discussion on cyberinfrastructure. One was "a robust and adaptable cyberinfrastructure that can link to other databases." A second feature highlighted was the ability to collect and use social science data with the physical, chemical, and biological data required to study environmental consequences of human activities. This feature would require data integration and modeling platforms capable of combining science and engineering databases and models with those of the social sciences where spatial and temporal and quantitative norms might be highly diverse. CI would need to be developed to span the interfacial areas and provide mechanisms to connect observations and create new platforms for knowledge creation.

Specific recommendations were provided on integrated data collection and storage. Data collection should be problem-oriented and focus on current problems and possible future problems. Therefore surveillance programs and data streams should have a quality assurance plan developed early on. Such a plan should link with the program and project objectives. The data must be up to the task of meeting CLEANER objectives for the WATERS Network. The match must be carefully evaluated at the initial program planning stage. A major challenge identified was how to control and record data quality collected in the field through numerous sensors and sampling devices. It was emphasized the quality of the data and its provenance produced by the observatories should be able to "identify principles and processes that are transferable to other locations."

The system to house and share data collected from the observatories must meet critical minimum requirements. A common framework is needed to link multiple databases that will become a "national resource" for hydrologic, environmental, and social data. National-scale environmental observing programs must be linked in order to coordinate and share resources that will

effectively "produce high quality data products for accurate scientific analysis and forecasting." A high level of system compatibility and integration would serve to benefit the "quality of the science and economics of system development and operation." The committee encouraged the continued integration of the approach adopted by CUAHSI's Hydrologic Information System (HIS) and the efforts to link the NEON CI and CLEANER CI with this system.

Key issues that are necessary for creating a common CI for environmental observatories are: 1) rules and guidelines for operation (policy); 2) standards for data, networks, and instruments; and 3) CI hardware and software. These were identified as essential components for developing "an efficient and secure, scalable and extensible" environmental observing system. These components must be carefully integrated with program goals and science objectives in the early planning stage of environmental observatories.

IV. CI Conceptual Design

The Waters CI conceptual design process takes as its starting point the results of the needs assessment above. A central role identified for CI is that it must enable collection and dissemination of data, data analysis and modeling through use of collaboration tools and the creation of a new kind of environmental scientist/engineer able to take advantage of all CI enables (NSF Workshop 2002 "role of CI"). Within the environmental field, the role in archiving and integrating data is clearly critical. The WATERS Network needs to be able to support environmental research that is observationally-oriented, highly collaborative and interdisciplinary, and generates and uses significant quantities of geospatially indexed data currently in a host of non-interoperable data formats. (NSF workshop 2002).

Thus, the WATERS Network is envisioned to include cyberinfrastructure consisting of (1) a virtual repository to archive existing and new data, (2) information technology to enable data modeling, analysis and visualization, and (3) a collaboratory enabling researchers to access and use the tools created in (1) and (2). Many additional aspects of CI will be considered in this report and indeed are critical to enabling these three aspects, but the top of **Figure 2**, focused on the needs of the community of researchers who will use CI to advance their domain-specific research will be the dominant perspective.

Figure 2. Conceptual framework of the components of CI and how they support the community of researchers

A. Scope

The primary goal of any cyberinfrastructure is to support the current and future science and engineering goals of the sponsoring community. The WATERS observatory community is defining a very challenging research agenda around national and global environmental issues and the cyberinfrastructure being envisioned must support this agenda. Specifically, from the survey results we know that:

- Data access and search of multiple data sources are perceived as the key contributions of CI;
- Researchers are willing to share data, but have concerns that they will not receive sufficient credit from others who use the data;
- Nonstandard and inconsistent data formats are the key barriers to data sharing;
- It is important for users to be able to identify data sources according to specific measures and variables of interest
- It is important for users to be able to understand the accuracy of data in order to gauge confidence in data sources;
- Hydrologists and environmental engineers have different needs when using data;
- Excel is the most popular application, followed by ArcGIS and MATLAB;
- Ease of use and utility are the drivers for technology adoption;
- Access to expertise, particularly complementary expertise, motivates the search for collaborators.

The challenge in building the WATERS CI is to meet the needs identified from the survey responses summarized above as well as the needs that follow from the science, social science, modeling, and education plans, while avoiding an over-engineered solution. That is, CI must be designed for both scientific and technological progress and focus on scalability and evolvability rather than aiming for an all-encompassing scale and scope. The WATERS design team must design and develop CI to assemble the resources required to address the issues driving the science and engineering plan yet recognize that the WATERS Network will not be the sole sensor network or observatory framework being created. Researchers will need a well managed set of core capabilities yet will also desire the ability to include new data sources and tools (from local projects or other national/international projects) to explore beyond the original science plan.

The sections below define the intended scope of the WATERS CI and explore the potential design space in terms of a representative use scenario, the required component framework, and cross-cutting concerns and design principles. While this discussion doesn’t conclude with a specific system design specification, it does identify key distinguishing features of the CI for the

WATERS Network and argue for an overall conceptual architecture that is well suited to the creation of an operational observatory that will be able to meet the goals of the WATERS science plan and serve as a catalyst for future research.

B. Use Scenario

To better understand how these components could work together to enable innovative science, education, and outreach, consider the following hypothetical scenario:

> *Evelyn Smith is a field researcher who studies hypoxia at a WATERS environmental field facility (EFF). Because hypoxia is a transient event, Evelyn relies on predictive models implemented within the WATERS collaboratory to notify her when the risk of hypoxia is high and she should enhance her field sampling. One hot summer day she gets an e-mail alert notifying her that hypoxic conditions are predicted with 80% certainty in 24 hours. She logs into the WATERS collaboratory where she joins an ongoing discussion with researchers (both local and across the country) who received the same alert and are looking at the current data and model predictions. The researchers agree that the predictions appear to be reasonable given the current conditions, and a number of researchers in the local region agree to mobilize their field crews and share their data. Evelyn notifies students and volunteers from the local region, who have previously indicated an interest in helping with field sampling at the EFF, that the field crews will be performing detailed manual sampling of the affected region the next morning.*
>
> *The next morning, data that the field crews collect are transmitted through the network back to the data store and model predictions are updated automatically. Additional data needs to address predictions with high uncertainty are identified and transmitted back to the crews through their collaboratory subscriptions. Around the country, other researchers, regulators, and students monitor real-time visualizations of the hypoxia event as it unfolds and discuss its implications in the collaboratory. Results are compared with those from previous events at other site.*
>
> *Joe Rodriguez previously searched for information on hypoxia in the collaboratory, so he receives a referral to check out the event next time he logs in. He's been creating a nationwide hypoxia data mining analysis, which he updates to identify common characteristics of recent events around the nation. He posts his results in the collaboratory and requests an online peer review. When the results receive a positive review, several regulators receive alerts, which they use to consider future regulatory options to reduce hypoxia.*

This type of scenario frames the required functionality in realistic terms focusing not only on what will be done, but how researchers will interact with the CI to accomplish their goals. Scenarios of this type are valuable both for communicating the potential benefits to researchers and educators as well as to specify end-to-end system requirements to developers.

C. Components

To enable this and similar types of large-scale, community-based research scenarios, and meet the needs identified in the initial drafts of the science plan, WATERS CI must be comprehensive

(including all processes related to the water cycle), operating in real (or near-real) time, and cover processes from the molecular to the watershed scale. Enabling this type of CI requires specific components. These fall into the three groups shown previously on **Figure 2.**

- Base Technology: Computation, Storage, Communication
    - High Performance Computing (HPC)
- Knowledge Management, Visualization, and Modeling
    - Metadata
    - Data Models
    - Data Standards
    - Modeling and Visualization[7]
- Networking, Operating Systems, and Middleware
    - Interoperability Service Oriented Architecture
    - Digital Libraries
- Collaboration Services

Details of these components are provided in Appendix F.

While not a focus of the WATERS CI committee (see the separate Sensors and Sensor Networks report), instruments that collect data and enable the observation of physical phenomena are a critical support component to the collaboration-facilitation envisioned by the WATERS plan. For example, one exciting possibility is that researchers might use the WATERS CI to teleoperate remote instrumentation.

D. Design Recommendations

The prerequisite for all other functionality of the WATERS Network observatory CI is the ability to capture data from the proposed sensor network, calibrate and apply quality control procedures to it, and provide ongoing access to it for the broad WATERS community. This leads to requirements in terms of networking, computational power, and data storage capabilities as well as software to support the overall data flow, to manage and monitor the system, and to provide effective data discovery and access. However, this set of requirements is not sufficient on its own to meet the larger goals of the observatory. To enable the development of new models of environmental processes and human impacts, the CI must support the incorporation of data directly into the full range of relevant models, the calibration of models, the comparison of

[7] This area is the focus of a separate WATERS committee report.

model predictions with each other and further observations (including feedback for adaptive observation), and the use of models for evaluating candidate designs and management practices. To serve as a community focus, the CI must support scientific discourse and support the large-scale dissemination of new ideas and resources. To serve as a model for future sensor networks and long-term, large-scale scientific data curation efforts, the CI must demonstrate best-practice capabilities that are independent of the specifics of environmental research and that can be cost-effectively adapted to changes in technologies and in scientific goals over decades.

The requirements that can be derived from these larger goals add significantly to the challenge of creating an effective CI for the WATERS Network, yet the committee firmly believes that they, and the core goals related to the capture and management of national-scale environmental observations, can be met with appropriate design. A number of cross-cutting concerns influence the applicability of potential design choices. These include:

- Interoperability
- Quality Control Assessment
- Scalability
- Distributed Content Management
- Distributed Process Management
- Virtual Organization-Based Management
- Provenance Management
- Open Interfaces and Protocols

Details of these concerns are provided in Appendix E.

E. Distinguishing CI for WATERS

The components of CI and the cross-cutting concerns described in sections B and C above are critical for development of WATERS CI. However, they are largely critical for the development of CI across many disciplines, and therefore are not unique to the WATERS Network or to environmental observatories more broadly.

The WATERS science plan has a strong emphasis on real-time modeling to support: (1) event-based data collection and experimental design (e.g., that adaptively responds to “hot moments” and “hot spots” to understand large-scale transient phenomena); (2) research to better understand the processes that lead to event-based phenomena; and, (3) testing of improved management

practices to better manage event-based phenomena. Addressing these specific goals from a CI perspective will create the following needs and issues[8]:

- *Access to large repositories of observational data.* Significant repositories of observational data already exist but long term preservation of these data is needed. Previous work by the CUAHSI HIS team had identified streamlined programmatic access to federal repositories of hydrologic observations as a critical need. Another challenge is engaging Federal agencies and other data collection stakeholders in data sharing. CUAHSI HIS is developing uniform web services and common vocabularies for hydrologic data, and for working with Federal agencies on data standards.

- *Large streaming datasets.* With numerous datasets streaming into the WATERS archive in real time that require rapid analysis and response, improved tools for handling large datasets are needed. Currently these datasets are handled in an ad hoc, time-consuming, manual process that will not scale well to real-time analysis by large numbers of researchers and educators.

- *On-demand high-performance computing (HPC).* Currently high-performance computing (HPC) is generally delivered in batch mode, where requests for computing time are submitted in a queue and results are delivered when time becomes available, which can be minutes, hours, days, or even weeks later. To support real-time, adaptive study of large-scale, transient phenomena, the WATERS community will need to run computationally-intensive models in real time, requiring the creation of new HPC modes of use. While other communities do need on-demand computing, they can generally make advance reservations for computing time, which will not be always be possible with transient environmental phenomena of interest to the WATERS community.

- *Improved access to HPC.* Currently very small numbers of researchers in environmental engineering and hydrology use HPC resources. Learning to use the resources is time-consuming and requires excellent programming skills that are beyond the reach of most students. Better tools to enable easier access to HPC are needed. Furthermore, models that are widely used in our community can be implemented in parallel on HPC once, and then shared with other researchers through an easy-to-use community modeling resource, thereby enabling broader use.

- *Distributed, interdisciplinary research support.* Implementation of the WATERS Network implies a substantial future increase in interdisciplinary research by teams of researchers across the country (i.e., “distributed research”). Working in this mode with large teams introduces significant challenges that many environmental engineering and hydrology researchers have not yet encountered. Better tools will be needed to support these teams so that they can work remotely efficiently and effectively.

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[8] These needs transcend environmental observatories as well, with other applications requiring these CI features. This list is not meat to be exclusive to WATERS or all-inclusive for WATERS needs; however, these are areas where WATERS could lead CI development.

In addition to these identified unique features, it is likely that research goals articulated in the science plan and expanded on by the researchers themselves will lead to specific CI needs. Thus, it is critical that the WATERS community gain a better understanding of specific CI that will be required to accomplish our science goals and the resources that will be needed to create such CI. This can be gained by (1) large-scale test-bedding[9] that begins to tackle end-to-end data to knowledge in a real observatory setting; (2) education to help the community understand what is currently possible with CI and what is needed to address observatory requirements; and, (3) extensive coordination with other planned observatories to better understand what CI development the WATERS Network can leverage from other efforts and what must be developed principally for the WATERS Network and then shared with other communities.

V. Conclusions

The success of research to address WATERS's grand challenges rests on the implementation of critical cyberinfrastructure. Without carefully planned, implemented, and maintained CI the research endeavors planned to address the urgent environmental questions of humanity will fail.

This report focuses on the evaluation of CI needs for the WATERS community and the development of a conceptual model of CI for environmental engineering and hydrology research and education.

Our evaluation of existing activities and coordination with other WATERS committees leads to the following conclusions.

1. Cyberinfrastructure potential and utility is still unclear within the community. More community education is needed, but more importantly, successful use of CI must be demonstrated on research projects of broad interest. Thus, funding for test-bedding must continue and be expanded.
2. Collaboration between CI domain experts and environmental engineering domain experts is critical for success. Cross-disciplinary research in test-bedding should be strongly encouraged.
3. Training students to work at the disciplinary boundaries between environmental engineering and computer science is a critical need that is poorly addressed by current educational paradigms.

Our consideration of similar programs leads to the following conclusions.

4. Many large scale CI development and deployment projects are slated to end within the next two years and a continuing stream of exploratory CI efforts across agencies is expected. It is critical that lessons learned in these projects are gathered and that the

[9] NSF's CEO:P program will initiate this type of test-bedding in FY07, but efforts that are focused on specific gaps in CI to address WATERS science needs will be needed in the future as the WATERS conceptual design becomes clear.

WATERS design team develops a means to coordinate and capture lessons-learned and technological advances from these programs.

5. Coordination among different NSF sponsored observatory programs is critically needed. We concur with the recommendation of the NRC committee that an organizing group with a mandate specifically to develop mechanism to coordinate standards across NSF observatories and other geospatial efforts is needed.

Our survey of the WATERS community leads to the following conclusions.

6. Echoing conclusion 1, the community expressed significant uncertainty regarding the potential and utility of CI.
7. Usability was a significant concern of the community, and WATERS design team must ensure that the design of its CI is clear and appealing, something with which users can readily identify. Further, sufficient support for training, maintenance and trouble-shooting must be available.
8. CI must be adaptable to the changing needs of the research and educational endeavors and to the developing technological sophistication of the users. What is needed and works well today may not meet the known or as-yet-unimagined needs of tomorrow.
9. Hydrologists and environmental engineers have different needs when using CI and thus both community-specific CI needs will have to be integrated for the network.
10. Among the many types of CI discussed, all survey respondents stressed the need for better access to data and improved methods to deal with data inconsistencies and data validity. Improved data management, including discovery, access, integration, analysis and visualization tools are a central concern for the WATERS Network and coherent access to current data and early access to new WATERS data are a high priority for the community.

This document also presents a set of concerns regarding the conceptual model of CI in the WATERS Network. The following conclusions are made:

11. The WATERS CI must be developed in close collaboration with observation programs that already exist and those that will be coming on line during the next decade and be architected to be able to leverage ongoing advances from the larger CI/IT community.

12. The WATERS CI must be architected for change: change in scale, change in component location, change in data types and community processes, change in scientific understanding, change in technologies, change in the community, and change in the overall scientific ecosystem. Further, the WATERS design team should use the cross-cutting concerns and design patterns outlined here as criteria for evaluating physical architectures and components.

13. Successful implementation of cyberinfrastructure (CI) for the WATERS Network will require rigorous validation and testing of applications and services to ensure ease of use and user acceptance.

14. Critical CI components are expected to be: high performance computing-based infrastructure; intensive database management for storage, access and sharing; data fusion and mining; model development and sharing for knowledge generation; visualization for value enhancement and communication; and collaboration and knowledge networking.

15. WATERS CI must link to existing federal hydrological research and monitoring programs that provide valuable databases that support the WATERS science plan and specific goals and research tasks.

Cyberinfrastructure that supports the WATERS vision will enable environmental engineering and hydrology education and research in the coming decades. This CI will be carefully planned, implemented, and maintained in collaboration with other observatory efforts. It will be responsive to the needs of the community, as articulated in this report. It will be extensible and adaptable to the needs that arise as the WATERS project moves from vision to implementation. And, finally, it will drive research and education activities to new horizons, enabling critical new questions about our environment to be asked and answered.

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Appendix A. Partial list of NSF Sponsored workshops and Reports on CI in Different Environmental Research Communities[10]

- March 2001 NEON Workshop II focused on CI. http://www.neoninc.org/documents/NEON2_Mar2000.pdf
- September 2002 NEON Workshop VI focused on CI. http://www.neoninc.org/documents/NEON6_Sept2002.pdf
- October 2002 NCAR sponsored a workshop on CI for Environmental Research and Education. http://www.ncar.ucar.edu/cyber/cyberreport.pdf
- Geoinformatics: Building Cyberinfrastructure for the Earth Sciences (2004); workshop held May 14 - 15, 2003; Kansas Geological Survey Report 2004-48
- http://www.geoinformatics.info/
- Trends in IT Infrastructure in the Ocean Sciences (2004); workshop held May 21-23, 2003, http://www.geo-prose.com/oceans_iti_trends/oceans_iti_trends_rpt.pdf
- June 2003 NSF-sponsored workshop on Cyberinfrastructure for Engineering Research and Education. http://www.nsf.gov/eng/general/Workshop/cyberinfrastructure/index.jsp
- August 2003 NSF sponsored a workshop on CI for Environmental sensors. http://www.lternet.edu/sensor_report/cyberRforWeb.pdf
- Geoscience Education and Cyberinfrastructure, Digital Library for Earth System Education, (2004); workshop held April 19-20, 2004, http://www.dlese.org/documents/reports/GeoEd-CI.pdf
- December 2001 - June 2004 - WATERS workshops; some included CI. http://WATERS.nacse.org/workshops/index.html
- Cyberinfrastructure for the Atmospheric Sciences in the 21st Century; workshop held June 2004 http://netstats.ucar.edu/cyrdas/report/cyrdas_report_final.pdf
- November 2004 NSF-sponsored workshop on Sensors for Environmental Observatories, which discussed sensor-related CI issues. http://www.nsf.gov/news/news_summ.jsp?cntn_id=105708&org=olpa&from=news
- December 2004 NSF Environmental Cyberinfrastructure Workshop discussed cross-cutting CI issues related to environmental observatories.
- March 2005: NSF SBE-CISE Workshop on Cyberinfrastructure and the Social Sciences. http://vis.sdsc.edu/sbe/
- Cyberlearning Workshop Series; workshops held Fall 2004 - Spring 2005 by the Computing Research Association (CRA) and the International Society of the Learning Sciences (ISLS)
- http://www.cra.org/Activities/workshops/cyberlearning
- October 2005: GEON CI workshop. http://geongrid.org/workshops/ciw_uofhyd/

[10] A broader list of CI workshops and reports is available at http://www.nsf.gov/od/oci/reports.jsp

- June 2005 NSF-sponsored workshop on Sensor Array Cyberinfrastructure and Informatics: Identifying Generic Solutions for Environmental Observatories. http://research.cens.ucla.edu/portal/page?_pageid=53,97468&_dad=portal&_schema=PORTAL
- May 2006 NSF-sponsored workshop on Modeling for Environmental Observatories discussed modeling-related CI issues that affect all proposed environmental observatories. http://www.modeling.uga.edu/tucson/index.html
- July 2006 GEON CI workshop. http://www.geongrid.org/workshops/beijing2006/

- NEON issued its Networking and Informatics Baseline Design (NIBD), which defines NEON's CI design, for the Conceptual Design Review held in November 2006. http://www.neoninc.org/documents/NIBD_2006Oct23.pdf

Appendix B. Summary of Review of Other Observatory CI plans

Consortium of Universities for the Advancement of Hydrologic Science (CUAHSI) Hydrologic Information Systems (HIS)

Phase I of the CUAHSI HIS project was supported by an NSF collaborative project (EAR-0413265) for the period January 2004 to March 2006 and extended to October 2006 by a supplement grant. The Phase I work produced a prototype HIS comprising **Web Services** that provide direct programmatic access to data from national databases, a **Hydrologic Data Access System** that provides a web browser interface into data served via CUAHSI web services, an **Observations Data Model** that provides a standard relational database schema for the storage of point observations from observatories and other experimental sites and a Digital Watershed Toolkit that contains tools for hydrologic modeling and analysis, developed in this project or contributed by others . These accomplishments are summarized in a status report (Maidment, 2005) and detailed in other reports and papers listed on the CUAHSI website under Hydrologic Information Systems (www.cuahsi.org/his.html).

Phase I of HIS focused on the conceptual design for data access of time series data published by Federal agencies and a strategy for publishing time series data from individual scientists using web services. Phase II of HIS will create operational products from those conceptual designs and develop a series of extensions to this design. There are five specific goals for Phase II

1. *Operational data discovery and publication services for point time-series data.* Current provisional web services must be made operational by finalizing designs for the map-based data access system, for web services to USGS, USEPA, and NCDC data holdings, and developing tools for academic scientists to publish their data using the Observations Data Model, a relational database schema for the storage of a broad class of observation data with sufficient ancillary information to allow it to be unambiguously interpreted and used. Web services providing access to the observations data model are included.

2. *Deployment of point data services to WATERS Test Beds*. This deployment will serve as a beta test of the operational designs. The WATERS Test Beds are a diverse set of projects which will identify how robust the current model is and will test the performance and reliability of the federated system. In addition, we will assess requirements of this user group for additional data streams (such as from NRCS SNOTEL).

3. *Development of Additional Data Services*. Beyond time series collected at a point, environmental scientists need access to static coverages (such as digital elevation models) and dynamic fields such as precipitation and remotely sensed data. There are many issues to be resolved before these data can be quickly and reliably delivered to the end-user in the same manner as point time series. Some design work for MODIS satellite products was done in Phase I, and the HIS project will coordinate with an ITR project that is working with NEXRAD precipitation products.

4. *Designing the Digital Observatory.* Once data are provided, it must be organized to permit analysis seamlessly across the four dimensions of space and time. Approaches for

data synthesis have been suggested, such as Water Information System, NetCDF, or ArcHydro. None of these are fully implemented or adequate for the digital observatory.

5. *Community Modeling Framework.* Within the digital observatory, multiple conceptualizations and multiple quantitative models will be supported. Again, some modeling environments have been developed, such as D2K, Kepler and OpenMI, but these environmental must be customized to work within the digital observatory. A guiding principle in this effort is the development of an open, extensible system that the community can contribute to.

The phase I HIS project has developed the architecture for HDAS and has developed web services to the provisional level (see below) for the following national databases.

- USGS National Water Information System (Daily Values, Groundwater, Real time unit values and instantaneous irregular data)
- DAYMET daily surface meteorology from the University of Montana, Numerical Terradynamic Simulation Group (NTSG)
- MODIS - Moderate Resolution Imaging Spectroradiometer instrument aboard the Terra (EOS AM) and Aqua (EOS PM) satellites
- NAM - NCEP North American Mesoscale (NAM) Weather Research and Forecasting (WRF) Model.
- EPA Storet water quality data

Some development web services are available for additional data sources and the CUAHSI observations data model. Prototype web services are available for data extraction and provision of metadata for the data sources indicated. Some data discovery web services are under development. During phase II the intention it to advance these web services to release status and develop the as not yet developed services, most notably those for data discovery and publication based on the observations data model.

In Phase 2 a careful evaluation process is needed to assign priorities to the products most in need of advancement and determining the best way of achieving that goal, either within the HIS team or by collaboration with the academic, commercial or government partners, who are keenly interested in the success of HIS.

The HIS team has determined that the priority **data access** items to be achieved during Year 1 of the Phase 2 effort (by Sept 30, 2007) are:

- **WaterOneFlow services** – carrying the federal water data web services from NWIS (Provisional), Storet (Development) and NCDC (Development) to a Release stage.
- **Observations Data Model** – carrying the current Observations Data Model structure (Provisional) and Data Loader (Development) to Release status by Sept 30, 2007.
- **Hydrologic Data Access System** – providing a robust map interface to the federal WaterOneFlow web services. This involves collaboration with a commercial partner, the

Environmental Systems Research Institute, which is donating programming resources to help CUAHSI achieve this goal.

- **HIS Workbook** – complete the implementation of tutorials and templates for ingesting the data from the federal water data web services into various application environments (Excel, ArcGIS, Matlab, programming languages).
- **Exploratory Assignments** – Interactions with the Test Bed principal investigators, CUAHSI members, and other potential users to determine desired capabilities in emerging areas (expanded below). Evaluation of what will need to be done to provide these capabilities by way of adding servers, expanding hardware for seamless interactions, and setting protocols for meta data development.

(1) Deployment to WATERS Test Beds

HIS will support observatories through the deployment of the WaterOneFlow system of web services and a local HDAS portal on a server hosted by an observatory. This cyberinfrastructure will provide the ability to construct "digital watersheds" as integrated assemblies of local and remote data sources. The **Observations Data Model (ODM)** provides a standard relational database schema for the storage and retrieval of hydrologic observations, other information needed to support desired interdisciplinary research, and sufficient ancillary information (metadata) about the data values to provide traceable heritage from raw measurements to usable information allowing them to be unambiguously interpreted and used. The ODM will be supported by web services for the discovery and publication of observatory data as well as tools that HIS will develop (such as the **Data Loader** and **ODM database tools)** for manipulation of ODM data and for initial data reduction and quality control. HIS will also support observatories by the establishment of digital watersheds. Digital watersheds comprise the **catalog** of national data specific to an observatory area of interest as well as **local data** both in an ODM as well as in formats that the ODM does not include, such as GIS layers. HIS will provide **sensor to database** capability to ingest information from data-logger files into the ODM in real time, i.e. as soon as the sensor data file is delivered from communications telemetry to the computer a service will run to ingest the file.

The NSF program in Hydrologic Science with support from CUAHSI and the NSF program in Environmental Engineering with support from the CLEANER Project Office are jointly pursuing establishment of the WATERS Observatory Network. These two NSF programs conducted a competition to establish “test-beds” to examine issues in the conceptual planning and implementation of the WATERS Network and 11 projects at 10 test-bed sites were selected (the Susquehanna Basin has two testbed projects with different goals and PI’s). Each testbed project will be dealing with different sorts of data and have different needs for data synthesis. HIS will have the responsibility for entering many types of information and supporting explanatory meta data into a common system. This service will define a general scope for data coverage by HIS. Phase II will involve planning and preliminary testing on a system providing this scope of coverage. Later phases in HIS will nurture the system to provide wider geographical coverage, add additional items, and make needed refinements to the hardware and software.

National Ecological Observatory Network (NEON)

NEON has a Committee on Information Technology and Communications that focuses on cyberinfrastructure needs for NEON. This committee released its first report in January 2005. NEON CI is viewed as transformational as it will provide (1) advanced collaborative environments, (2) taskable sensors networks, and (3) efficient support for model development and integration.

The first report of this committee (Jan 2005) focused on the use of a web services model for cyberinfrastructure. One objective was to develop workflow execution tools to "enable users to bring processes together as required for particular applications." This report also included plans to leverage on-going CI efforts throughout NSF. The second report of this committee (March 2005) focused on a structure for the sensing systems within the observatories. Motes were defined as distributed small nodes while larger microservers were considered larger nodes. The vision of the collective system is that it will be taskable and adaptive, including autonomous features.

In October 2006, NEON released a formal report of the activities of this committee, the Networking and Informatics Baseline Design for the National Ecological Observatory Network[11]. This report provides significant information on the CI plans for NEON. The plan focuses on an 'end-to-end' digital solution for NEON data. Five key functions are envisioned for NEON CI:

1. Embedded cyberinfrastructure for distributed instrument control and reliable data transport,
2. Data curation and archiving,
3. Data analysis, integration, modeling, and visualization,
4. NEON portals, and
5. Collaboration environments.

Five principles are described to guide NEON CI development:

1. Provide open access to data and information,
2. Employ an open architecture, in developing a robust, high performance infrastructure,
3. Engage NEON stakeholders in cyberinfrastructure design and operation,
4. Optimally leverage with partnering organizations,
5. Design for modularity, security, simplicity, and standardization.

[11] Available on the web at http://www.neoninc.org/documents/NIBD_2006Oct23.pdf.

The Ocean Research Interactive Observatory Networks (ORION) and the Ocean Observatories Initiative (OOI)

In August 2006, ORION held a conceptual design meeting that led to a report entitled, The ORION Cyberinfrastructure Conceptual Architecture.[12] The OOI is envisioned as an activity that will transform ocean sciences from "an individual, short-term expeditionary mode to a persistent, collaborative observation mode." A major focus of this report is the need within observatories for real-time access and response to detected events. This requires architecture described as a "knowledge network, a distributed collection of information that can be accessed in a coherent way."

The OOI report lists core characteristics of the planned CI architecture as:

1. ORION is the federation of independently operated facilities that subscribe to a common operating model expressed as a common set of service protocols.
2. The scope of ORION functionality is defined by the aggregate portfolio of service protocols provided by its participating facilities.
3. The role and scale of any participating facility are determined by the service protocols it chooses to present and the resources it chooses to commit.
4. The provision of services from a participating facility will be monitored for quality and conformance to protocols.
5. A facility may delegate the authority, operations, and capabilities of a resource to any other facility.
6. Machine-to-machine interactions are given comparable stature to human-to-machine interactions.

A common operating model is proposed that incorporates observatory components all the way to the level of the individual sensors. Additional concerns to be addressed by the CI planning group include:

- Incorporation of real-time streaming data into the scientific process
- Specification of common instrument services applicable to all ORION sensors
- Construction of a system of systems that is scalable, extensible, and manageable
- Interfacing the ORION CI and resources with external organizations and resources.

[12] The executive summary of this report is available online at http://www.orionprogram.org/organization/committees/ciarch/CI_ExecSumm.pdf.

Appendix C. CLEANER[13] End-Users Survey

Methods

Sample Selection and Composition

The survey sample was derived from the members of the Association of Environmental Engineering and Science Professors (AEESP)—composed of professors in academic programs who provide education in the sciences and technologies of environmental protection—and the Consortium of Universities for the Advancement of Hydrologic Science, Inc. (CUAHSI)—a community of universities developing infrastructure and services to advance hydrologic science and education in the United States. These two communities served as proxies for the environmental engineering and hydrologic science communities. We selected 300 members from each group, representing a random, stratified sample proportional to the geographic distribution of the membership across the United States. We also cross-examined the sample lists to ensure that no one was listed on both lists.

We received 313 responses, representing a response rate of 52%. Of these responses, 47% were from AEESP and 53% from CUAHSI. Response rates were also proportionally similar to the sample selection. The respondents were 82% male and 8% female (the remainder did not answer this question). Most respondents had PhD degrees (95%), and the median number of years since their highest degree was 15 years (mean number of years = 17). Almost all of the respondents were unfamiliar with the CyberCollaboratory; 64% of respondents never had heard of it before the survey, and 28% had heard of it but had not used it. While 53% of respondents were familiar with the CUAHSI project (as one might expect given the proportion of association members surveyed), only 33% were familiar with the CLEANER project. Among the respondents who were not familiar with the CLEANER project, nearly half of all respondents had heard of it but were not familiar with it.

Survey Design and Content

We designed the survey to capture the priorities and requirements of the potential target users. The questions covered familiarity with CLEANER and CUAHSI projects, knowledge about cyberinfrastructure, data use, technology adoption, attitudes toward collaboration, and potential use of a referral system. We also asked open-ended questions about their interest in using CyberCollaboratory and feedback for the CLEANER and CUAHSI projects. To ensure comprehensive coverage of technical issues and identify other important topics, we reviewed User Needs Chapter in the CUAHSI's Hydrologic Information System Status Report (Bandaragoda, Tarboton, and Maidment, 2005) and a pilot survey conducted for one of the CLEANER planning grants (Bhagwat, 2005). We pre-tested our survey with naïve users and

[13] At the time the survey was administered the term "WATERS Network" did not exist and all survey questions referred specifically to CLEANER. The CLEANER project was officially renamed as the WATERS Network in early 2006.

subject experts to ensure that our survey questions would be clearly understood and easily answered by our respondents.

Administration of Survey

The survey was administered during January and February 2006. The invitation to take the survey contained endorsement letters from the CLEANER[14] project office and the CUAHSI director. With these endorsements, we included a cover letter explaining the nature of the study, including a URL for accessing the survey. The survey was administered online using SurveyMonkey.com. The letters contained unique identification numbers that enabled us to track who had responded to the survey. We also enclosed a $2 bill as a cash pre-incentive (Birnholtz et al., 2003; Church, 1993). Approximately one week after mailing the letters, we sent reminders to everyone who had not yet responded. Two more reminder e-mails containing a link to the survey were sent to those in the entire sample who had not yet completed the survey. The reminders also contained unique identifiers allowing us to track who had responded to the survey.

Results

The following results are grouped according to the three main topic areas: data, technology use and adoption, and collaboration. In this section we report the quantitative results, and in the discussion, we will illuminate our findings with selected responses to the open-ended questions. The results are often presented in aggregate form, but where relevant, we have also identified notable differences between the AEESP and CUAHSI respondents, listing what percentage of total respondents were from each group. Figures displaying exact percentages are provided at the end of this appendix.

Data

The main questions specifically targeted to data use were about features of new data sources, specific databases, and obstacles to using data from different sources. Both groups agreed that the most important features of new data sources were finding specific measures or variables (57%) and the accuracy of the data (52%). However, the survey results show that other priorities were somewhat different. For example, the third most important feature was specific geographic locations (30% of the total sample selected this item), but the CUAHSI members represented 73% of those interested compared to 27% of those affiliated with AEESP. The time frame of data was relatively more important to AEESP members, who represented 44% of those who selected that feature.

We provided an extensive list of common data sources provided by government agencies. The USGS streamflow database is the primary data source overall (46%). The next most important data sources overall include United States Geological Survey (USGS) water quality (National Water Information System [NWIS], National Water-Quality Assessment [NAWQA]), Environmental Protection Agency (EPA) water quality (storage and retrieval repository

[14] The CLEANER Project Office has since become the WATERS Network Project Office

[STORET]), and digital elevation models (e.g., National Elevation Dataset and derivatives). However, each group has very different needs for data sources. For example, CUAHSI members are substantially more reliant on USGS streamflow data than AEESP members. (Of the respondents who selected this item, 72% were CUAHSI members versus 28% AEESP members.) Instead, AEESP members had a substantially greater interest than CUAHSI members in the USGS and EPA water quality databases, representing 63% and 78% of those reliant on those two data sources. Also, digital elevation models were much more important for CUAHSI members, who represented 87% of those who selected this data source.

Finally, we asked respondents to identify the obstacles to using data from different sources. Cumulatively, nearly one third of respondents felt that nonstandard or inconsistent data formats or units were the most important obstacles. This includes nonstandard formats or spatial scales, unknown units, and irregular time steps. Learning how to quality control the data was the next most significant obstacle (21%) and the number of respondents who identified metadata problems as an important obstacle to using data from different sources was also small (14%).

Given that many government funding sources require that data be made available after a specified amount of time, it was not surprising that most respondents were willing to share data with others (77%); however, many had concerns about sharing data. In particular, 55% agreed that they were concerned that others would use their data without giving them sufficient credit.

Technology Use and Adoption

To understand respondents' technology use and adoption, we asked them what technologies they use and what would influence them to adopt or not adopt a new technology. Among common software packages used in environmental engineering and hydrology research, Excel was the most popular software for both groups by far, with 88% of total respondents identifying it as software they use frequently. The next most commonly used software packages were ArcGIS (37%) and MATLAB (33%). All other software use was among small subsets of the sample. These included SAS, MS Access, SPSS, MS Word, MS PowerPoint, SigmaPlot, PHREEQC, and FORTRAN, among others.

We were particularly concerned with what would be the hurdles or incentives for adopting new technology. The most important factors in adoption of new technology were clarity of interface and ease of use (49%) as well as the ability to do things they cannot do with current software or hardware (44%). Professional technical support (35%) and software that fulfills current research needs (32%) were more modest priorities. Respondents were less concerned about tedious tasks, such as the need to learn new tools (12%), having to install software on their personal computers (5%) or the necessity of creating an account (2%), when adopting a new technology. This suggests that researchers in these groups are willing to tolerate some inconvenience in return for the possible benefits.

We offered a list of possible CI resources and asked respondents to identify the three resources most useful to them for their research. Accessing data sources (46%) and the ability to search multiple data sources in one place (43%) were the two most useful CI resources. The other useful CI resources mostly concerned aspects of data manipulation, such as data visualization (33%),

conversion of data to a common format (26%), and data mining (21%). Although CI can provide advanced computational features, resources such as high-performance computing, workflow tools for setting up a sequence of data acquisition and analysis services, and storage of data and computations at another location were perceived as most useful to only a few respondents. Considering that 44% of respondents have indicated that the ability to do things they cannot do with current software or hardware is an important factor in adopting a new technology, this suggests that most may not be fully aware of the major new capabilities of CI such as workflows and data analysis with large data sets.

Collaboration

To calibrate our understanding of how respondents might use CI to support collaboration, we asked them how difficult it was for them to find potential collaborators and what would be compelling reasons to collaborate. We also asked how they might use CI to support collaboration, including several questions that gauged how they might respond to a system that uses social network analysis to provide recommendations (much as Amazon.com recommends books purchased by others).

Very few people expressed having a problem with locating collaborators. About half of the respondents felt that they did not find it difficult to locate potential collaborators who share their interests or who could provide different expertise from their own. Less than 20% found it difficult to locate collaborators, and about 30% were neutral on this topic. However, we divided the sample into three cohorts: people who graduated no more than 10 years ago, people who graduated more than 20 years ago, and those in between. A Chi-square distribution test comparing these cohorts indicated that more recent graduates found it more difficult to locate collaborators with *different* expertise than their own, and this difference was statistically significant, but there was no apparent cohort difference for identifying potential collaborators who share their interests.

When asked what were the most compelling factors that would lead them to collaborate, respondents identified access to complementary areas of expertise (54%) and access to another's expertise more generally (50%). Secondary factors prompting collaboration included shared interests (39%), having an opportunity to brainstorm ideas with others (27%), trusting a potential collaborator (27%), access to equipment (19%), and access to data (18%). Thus, when asked what CI capabilities that could support collaboration would be most useful, the simple ability to collaborate with others (45%), share data with others (43%), and have access to experts (36%) stood out. Curiously, these capabilities are more abstract, compared to other very specific capabilities such as tools for videoconferencing, collaborative editing, or scheduling meetings, which attracted very few people.

In order to identify how a referral system would be appropriate for this community, we suggested that we could make a diagram available that would show their connections to others in the system. Then we asked whether they would mind if a publicly available diagram showed that they were peripheral, somewhat connected to the network, or highly connected to the network. Most respondents were not concerned about such a display, regardless of how well they were connected to the network (a range of 71% to 87% would not mind), but as might be expected,

people minded diagrammatic displays more if they were less well connected to the network. Referral networks such as these also can provide a variety of information about the members and their activities. Respondents felt fairly comfortable with having personal information, such as their areas of expertise (84%), institutional affiliations (77%), and project collaborations (69%), available to others. However, they were more mixed about providing their demographic and contact information (56%). Unfortunately, we did not ask about demographic information separately from contact information, and some indicated that demographic data was more problematic than contact data. Given that most people can be found on a departmental website, this makes sense. What was particularly remarkable was that only 36% were fine with making records of their use of resources available to others.

Conclusions

Data

An initial issue for deployment of CI will be how to prioritize access to resources given that environmental engineers and hydrologic scientists favor different databases for their research. Moreover, search tools will need to allow flexibility within the search function so that researchers can find data based on specific measures or variables of interest, whether they are geographic, time based, or some other metric. Quality control is, of course, another huge issue, and systems will need to offer metadata that allows scientists to assess data and determine how they might use it with their existing data. As one of the respondents commented, “As a modeler, my major concern is often the difficulty of finding the quality of data that I need.”

Data sharing raises many issues. Some people simply want ready access to data without necessarily collaborating with another group. For example, one respondent said, “I would be most interested in obtaining easy and reliable access to field experimental data, mainly for the purpose of testing my group’s theories and models against these data.” However, data collected within the sensor network or posted to the storage spaces provided by the CI needs to be guarded to some extent. As one person remarked, “I would not want to give up my data the second I collected it and before I had a chance to analyze it and publish based on it. After I had a first crack at my data, then sure, others could have it.” Likewise, another said, “Sharing data and credit for use would depend on whether I had rights to my own data before I shared it with others.” In particular, people worry that fair credit be given for data provided to others. Credit issues could seriously compromise the incentives to contribute to a CI resource. One researcher said, “I would spend time considering a clear data and assignment of credit policy consistent with academic norms.”

Technology Use and Adoption

Although backwards compatibility with popular software applications might initially seem a large hurdle, Excel appears to be the most important tool that must be considered. However, a good number of respondents do want to go beyond current capabilities with the technology. This might not necessarily mean that they want to do fancier things, but simply that they want to accomplish what they already do in a better way. One person explained, “I am interested if this environment helps me carry out my research in a more efficient and effective manner.” Clearly

many people would be satisfied by simply having access to data through one interface. The challenge is convincing community members that these tools are easy to use and incorporate in their regular routine. Given that nearly half of the respondents thought that clarity of interface is an important obstacle to or incentive for adoption, developers will need to go substantially beyond "proof of concept" software, to providing a well-designed application that includes some degree of professional technical support. Researchers recognize the ephemeral quality of NSF-funded initiatives and the lack of funding toward supporting and maintaining tangible resources. As one said, "I have seen previous efforts like this to provide access to significant data sets etc. fail because the institutional support disappeared after the funding ran out." Thus, the relationship between offering a new technology and having it accepted is tenuous. As one respondent's remarks may speak for many others: "I'd be willing to try it, however, my evaluation period would be short: 3 months and if I did not receive benefit within that time I would probably drop it."

Collaboration

A good number of people are interested in the potential for collaboration that CI might offer them. In particular, CI can bridge distances between disciplines or universities and colleges. One respondent was particularly interested with respect to its ability to facilitate interdisciplinary research among top scientists, irrespective of their locations. Another remarked on the equality it offers: "I am very interested. I think it provides opportunities for faculty in smaller programs to participate in collaborative efforts." Our results suggest that finding people with *different* expertise is the most compelling benefit, but people with complementary resources and ideas are also appealing. To this end, referral systems could be quite useful, but they may need to consider how to support a person's search for collaborators who fill the gaps in their own expertise. Typically, referral and recommender systems are designed to show people and resources with something in common. These needs do not, however, point to a need for CI to offer fancier communication and collaboration systems or to duplicate existing capabilities. For example, synchronous and asynchronous communication tools are less of a priority than simply finding people and sharing files and ideas.

Finally, the open-ended questions identified an unexpected obstacle to participation: the perception that the community created by CLEANER (*and thus the WATERS Network*) would not be inclusive or welcoming. For example, one respondent said, "My biggest concern is that these are 'closed' groups and not open to new collaborations." Another remarked more generally, "So far the CLEANER Community is closed one. It will not succeed without opening it to others." Yet another reinforced this sentiment, "I am disappointed in the lack of opportunity to participate in this process." Perhaps part of the problem is the generally low profile of the CLEANER and CUAHSI initiatives across the wider environmental engineering and hydrology community. As a possible remedy, one respondent suggested, "Perhaps a broader effort to increase the awareness of these projects would be a valuable task."

Figure 1: Familiarity with Projects

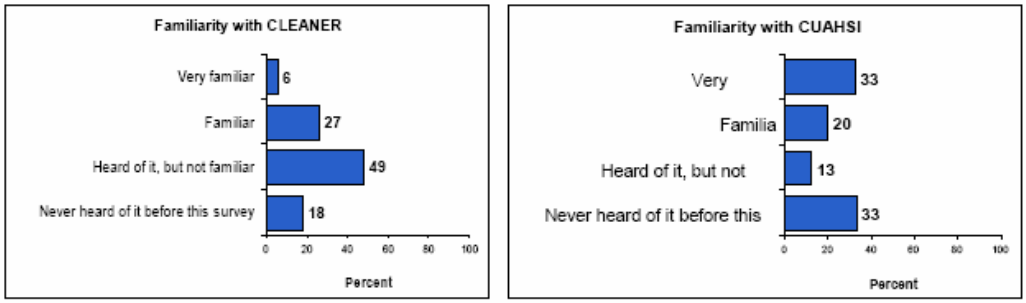


Figure 2: Important Features of New Data Sources

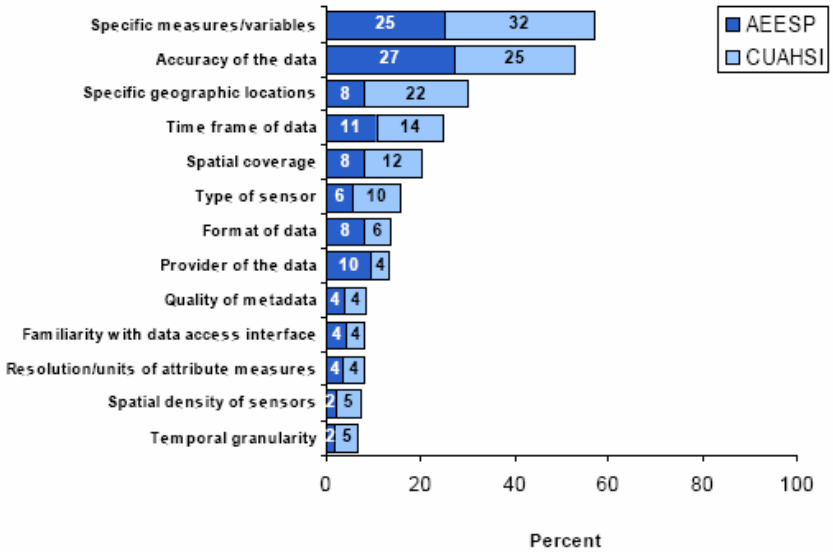


Figure 3: Current Important Data Sources

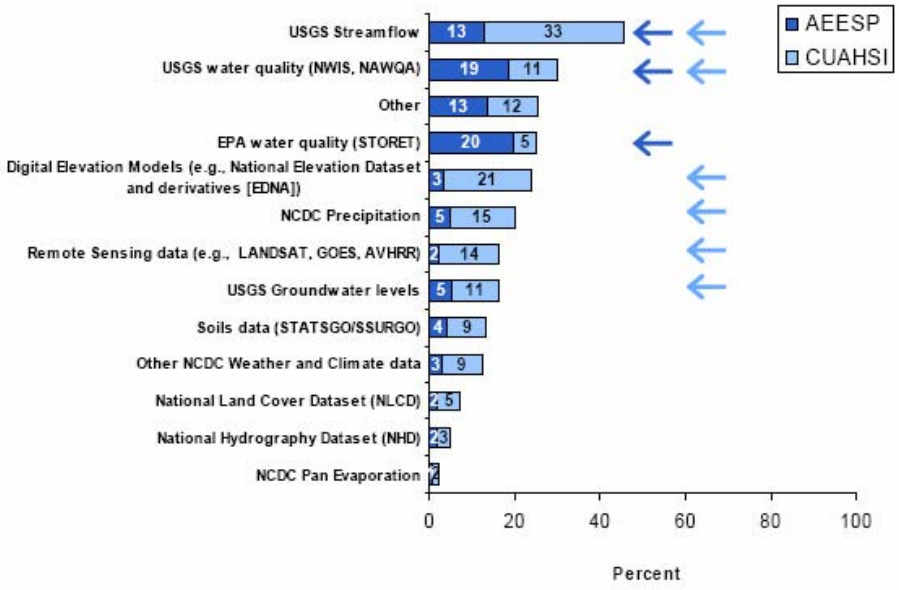


Figure 4: Most Important Obstacles to Using Data from Different Sources

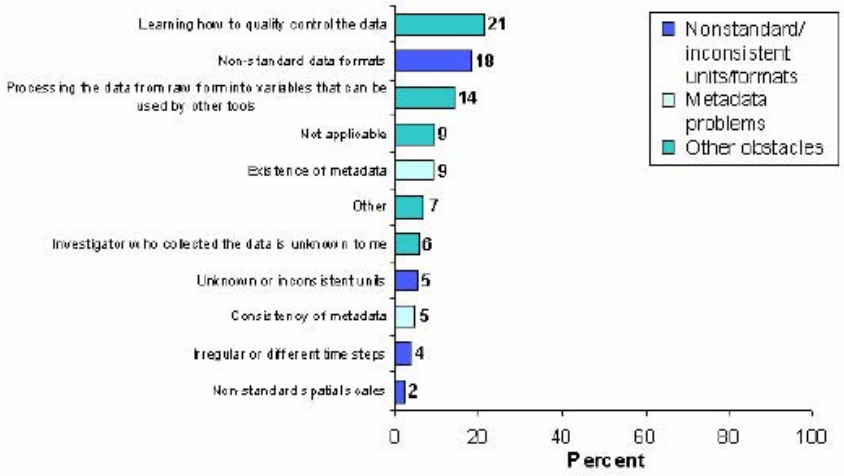


Figure 5: Popular Software Packages

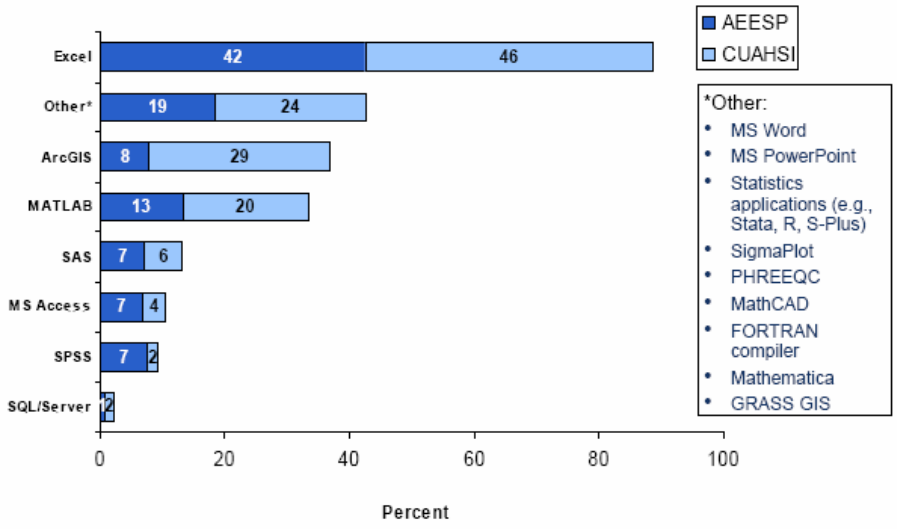


Figure 6: Factors Influencing Technology Adoption

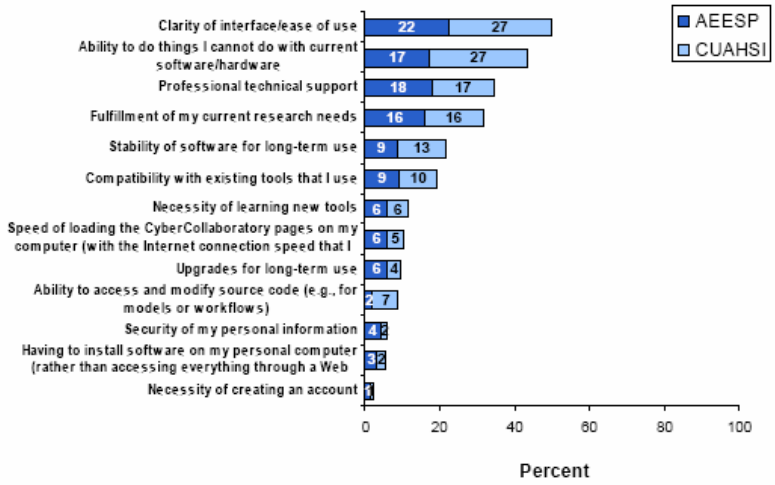


Figure 7: Interest in CI Resources

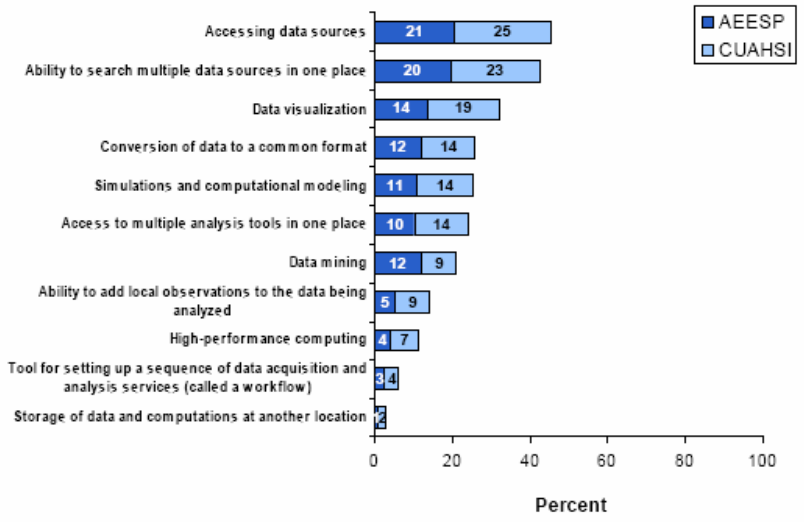


Figure 8: Difficulty with Locating Potential Collaborators

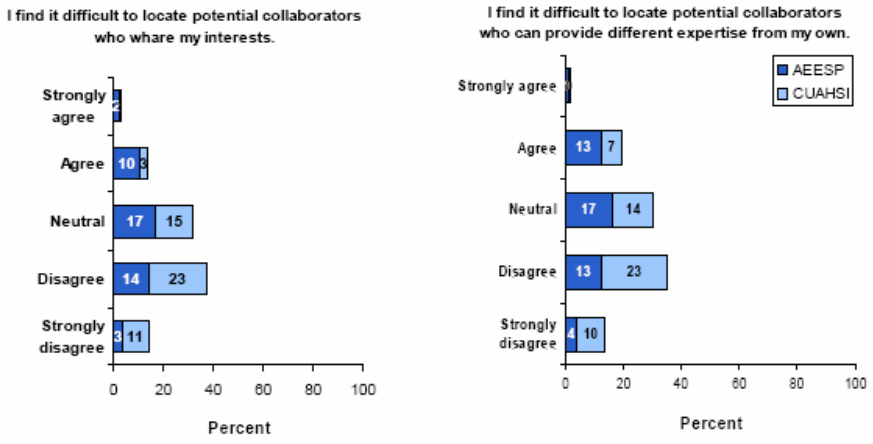


Figure 9: Compelling Factors of Collaboration

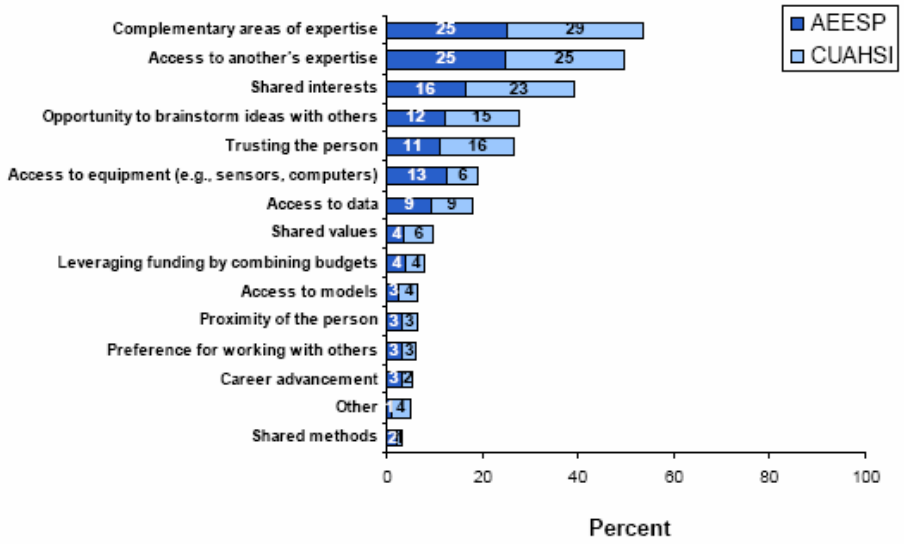


Figure 10: Useful CI Capabilities for Collaboration

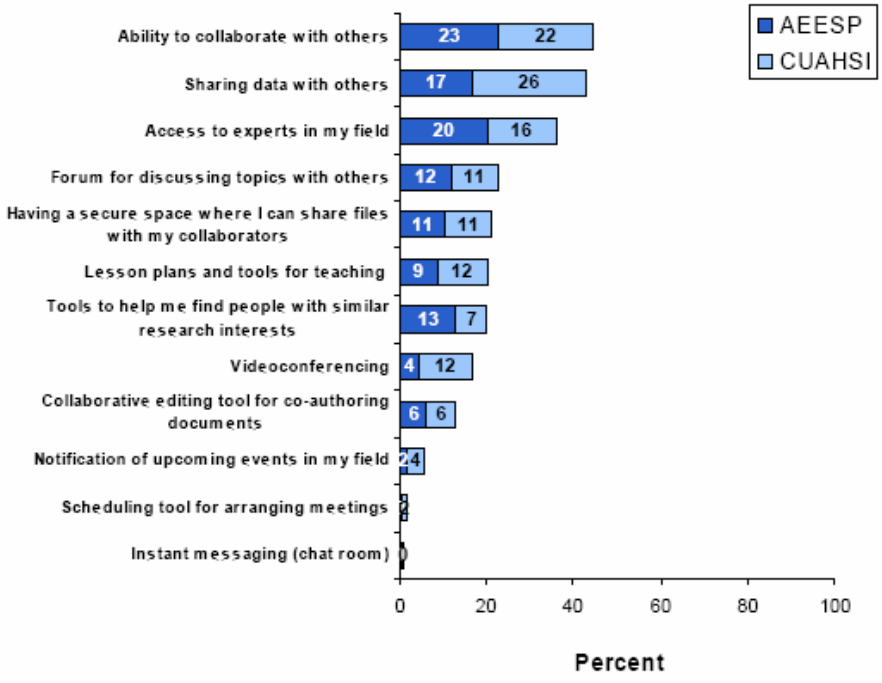


Figure 11: Networking Capabilities

Question: would you mind if this diagram were made available for other to view if you were…

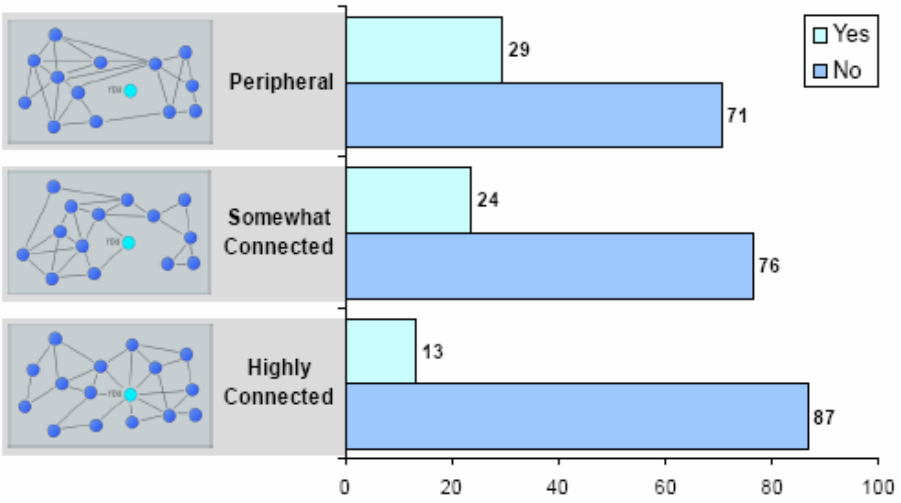
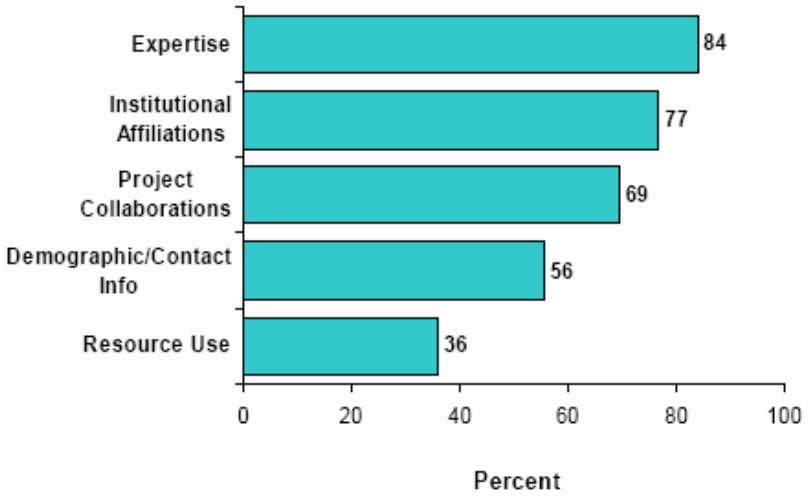


Figure 12: Networking Capabilities

Question: I would feel fine if the following information about me were available to others.



Survey Questions

1. How would you describe the kind of work that you do? (e.g., snow hydrology, coastal margins modeling, etc.)

Open-Ended Response

2. How familiar are you with NSF's proposed CLEANER (Collaborative Large-scale Engineering Analysis Network for Environmental Research) project?

Never heard of it before this survey

Heard of it, but not familiar

Familiar

Very familiar

3. How familiar are you with CUAHSI (Consortium of Universities for the Advancement of Hydrologic Science, Inc.)?

Never heard of it before this survey

Heard of it, but not familiar

Familiar

Very familiar

4. In a typical week, how many hours do you spend doing each of the following: Hours doing research-related work:

Hours doing teaching (including grading, meeting with students, and class prep):

Hours doing administrative work:

Hours doing consulting:

Other (Please specify task and hours):

5. Which one of these is your most time consuming research activity?

Collection of data

Locating data sources

Data reduction

Data visualization

Quality control of data

Coding numerical models or simulations

Troubleshooting computer code or unexpected results

Running numerical models or simulations

Theoretical work

Developing of new sensors, tools or techniques

Training and education

Writing papers and proposals

Project management and collaboration

Finding people with complementary expertise

Other

6. What is the number of hours you devote to this activity in a typical week?

Open-Ended Response

7. When you look for new data sources, what three features are the most important? (please select three)

Type of sensor

Format of data

Specific measures/variables

Specific geographic locations

Provider of the data

Time frame of data

Spatial density of sensors

Spatial coverage

Temporal granularity

Resolution/units of attribute measures

Accuracy of the data

Quality of metadata

Familiarity with data access interface

8. Which of the following data sources are the most important in your work? (please select three)

USGS Streamflow

USGS Groundwater levels

USGS water quality (NWIS, NAWQA)

EPA water quality (STORET)

NCDC Precipitation

NCDC Pan Evaporation

Other NCDC Weather and Climate data

Remote Sensing data (e.g., LANDSAT, GOES, AVHRR)

Digital Elevation Models (e.g., National Elevation Dataset and derivatives [EDNA])

National Land Cover Dataset (NLCD)

Soils data (STATSGO/SSURGO)

National Hydrography Dataset (NHD)

Other

9. What are the three software packages you use most frequently in your work?

Excel

MS Access

SQL/Server

ArcGIS

MATLAB

SAS

SPSS

Other

10. What environmental or hydrologic model do you use most in your research?

Open-Ended Response

11. In general, when you have collected data on your own or with a group of colleagues, how willing are you to share that data with other individuals or groups?

Very unwilling

Unwilling

Somewhat willing

Willing

Very willing

Not applicable

12. In your experience, what is the single most important obstacle to using data from different sources? (please select one)

Existence of metadata

Consistency of metadata

Non-standard data formats

Non-standard spatial scales

Non-standard spatial extent

Unknown or inconsistent units

Irregular or different time steps

Investigator who collected the data is unknown to me

Learning how to quality control the data

Processing the data from raw form into variables that can be used by other tools

Not applicable

Other

13. Please indicate your level of agreement with this statement: I find it difficult to locate potential collaborators who share my interests.

Strongly disagree

Disagree

Neutral

Agree

Strongly agree

14. What are the three most compelling factors that would lead you to collaborate with another person in your field?

Access to data

Access to models

Access to another's expertise

Access to equipment (e.g., sensors, computers)

Complementary areas of expertise

Preference for working with others

Opportunity to brainstorm ideas with others

Proximity of the person

Shared interests

Shared methods

Shared values

Trusting the person

Career advancement

Leveraging funding by combining budgets

Other

For Questions 15 and 16: Cyberinfrastructure (CI) is the high-performance network of coordinated technology resources (comprised of software, hardware, non-computer technologies, and human expertise) that provide advanced computational, collaborative,

data acquisition, and management services.

15. Based on your needs for your research work, please select the three cyberinfrastructure resources that would be most useful for you.

Accessing data sources

Ability to search multiple data sources in one place

Access to multiple analysis tools in one place

Ability to add local observations to the data being analyzed

Conversion of data to a common format

Data mining

Data visualization

Simulations and computational modeling

High-performance computing

Tool for setting up a sequence of data acquisition and analysis services (called a 'workflow')

Storage of data and computations at another location

16. Based on your needs for your research work, please select the three cyberinfrastructure capabilities or features for collaboration that would be most useful for you.

Sharing data with others

Having a secure space where I can share files with my collaborators

Access to experts in my field

Tools to help me find people with similar research interests

Ability to collaborate with others

Forum for discussing topics with others

Scheduling tool for arranging meetings

Videoconferencing

Instant messaging (chat room)

Collaborative editing tool for co-authoring documents

Lesson plans and tools for teaching

Notification of upcoming events in my field

For Questions 17 and 18: The CLEANER-CUAHSI CyberCollaboratory is a cyberinfrastructure environment envisioned to offer an array of tools for conducting research.

17. Please choose the level that best describes your use of the prototype CyberCollaboratory.

I have never even heard of the CyberCollaboratory before this survey.

I have heard of the CyberCollaboratory, but I have not used it.

I tried the CyberCollaboratory once.

I used the CyberCollaboratory a few times

I have used the CyberCollaboratory weekly.

I have used the CyberCollaboratory daily.

18. Which of the following factors would be the most important (as a benefit or hindrance) in terms of whether you would choose to adopt a new technology such as the CyberCollaboratory? (please select three)

Speed of loading the CyberCollaboratory pages on my computer (with the Internet connection speed that I have)

Stability of software for long-term use

Upgrades for long-term use

Ability to access and modify source code (e.g., for models or workflows)

Professional technical support

Clarity of interface/ease of use

Ability to do things I cannot do with current software/hardware

Necessity of learning new tools

Compatibility with existing tools that I use

Having to install software on my personal computer (rather than accessing everything through a Web interface)

Necessity of creating an account

Security of my personal information

Fulfillment of my current research needs

19. Please indicate your level of agreement with this statement: I find it difficult to locate potential collaborators who can provide different expertise from my own.

Strongly disagree

Disagree

Neutral

Agree

Strongly agree

Consider a diagram representing the connection of you and your research interests to others in the community. Would you mind if this diagram were made available for others to view if you were:

20. Peripheral to the network (one or no connections)?

Yes

No

21. Somewhat connected to the network?

Yes

No

22. Highly connected to the network?

Yes

No

The following six items apply to information collected and provided within the context of the CyberCollaboratory to members of the community. Please indicate your level of agreement with each statement.

23. I would be very concerned if my institutional affiliations were available to others.

Strongly disagree

Disagree

Neutral

Agree

Strongly agree

24. I would feel very fine if my project collaborations were available to others.

Strongly disagree

Disagree

Neutral

Agree

Strongly agree

25. It would bother me very much if records of my use of resources (e.g., datasets, software) on the CyberCollaboratory were made available to others.

Strongly disagree

Disagree

Neutral

Agree

Strongly agree

26. I would be very pleased if descriptions of my areas of expertise were available to others.

Strongly disagree

Disagree

Neutral

Agree

Strongly agree

27. I would mind very much if my demographic and contact information were available to others.

Strongly disagree

Disagree

Neutral

Agree

Strongly agree

28. I am very concerned that others would use my data without giving me sufficient credit.

Strongly disagree

Disagree

Neutral

Agree

Strongly agree

29. How interested are you in using a cyberinfrastructure environment like the CyberCollaboratory and why?

Open-Ended Response

30. Please provide any other comments that you would like the CLEANER and CUAHSI planning committees to consider for their planning purposes.

Open-Ended Response

These answers will remain confidential and secure. All personal information will be presented only in an aggregate form in reports and publications.

31. Your institutional affiliation:

Open-Ended Response

32. Job title:

Open-Ended Response

33. What is your highest degree?

BS or BA

MS or MA

Ph.D. or equivalent

Other

34. In what year did you receive your highest degree?

Open-Ended Response

35. Your gender

Male

Female

Follow-up Questions

Your answers to the following questions will NOT be connected with the rest of your survey responses.

1. Would you be interested in trying out the CyberCollaboratory as functional products become available for testing?

Yes

No, thanks

2. Would you be willing to participate in a 15-20 minute telephone or in-person interview to aid in the interpretation of these research results? (Note: 20-30 persons out of 600 total participants will be selected to participate in these interviews. Answering "yes" to this item does not necessarily mean you will be selected.)

Yes

No

3. Would you like to receive information about the results of this survey?

Yes

No

4. If you answered "yes" to any of these questions, please provide an email address at which we can contact you. (Your email will be used for no other purposes than those indicated above.):

Open-Ended Response

Appendix D. Review of CLEANER Planning Grants

Collaborative Project-Riverscope: Large-Scale Engineering Analysis Network for Environmental Research on the Hudson River [Robin Bell (Columbia University), Arthur Sanderson (Rensselaer Polytechnic University)]

This project defined CI as the interface between environmental sensors and communication systems (hardwired, static and wireless devices). Key elements of the CI include sensor networks, telecommunications, high performance and grid computing, real-time systems, database systems and protocols, visualization, data mining, related protocols and standards. The researchers considered CI to form the essential linkage between the Environmental Field Facility (EFF) and the Environmental Analysis Network (EAN). This project was undertaken in conjunction with "Hudson Riverscope," a joint project of the Lamont-Doherty Earth Observatory, Columbia University, and Rensselaer that was funded in 2000 by NY State and federal partners. The Hudson Riverscope includes monitoring sites on the Hudson River in Waterford, Saratoga County, and Piermont Pier, Rockland County, that provide data on the flow of materials in the river, the spread of zebra mussel larvae, and tracking of suspended contaminants such as polychlorinated biphenyls (PCBs). The project has predominately focused to date on deployment and testing of the sensing technologies; however, some work on algorithms for autonomous adaptive sampling and deployment optimization have been developed. Further, communication system development has been incorporated into this work, with wireless technology playing a significant role. A cyberinfrastructure testbed (CIT) was planned but has not been developed yet. It is not clear that the CI development or its structure has significantly limited the research progress. No CI problems or challenges were noted and no additional CI needs were identified during implementation of this project.

Collaborative Proposal (WATERS): Cyberinfrastructure Needs for a Model Environmental Field Facility in Baltimore, Maryland as Part of an Engineering Analysis Network [Claire Welty and Michael P. McGuire (University of Maryland, Baltimore County), Michael Piasecki (Drexel University)]

In this project, CI is defined as the data handling, analysis, visualization, sharing, and modeling from a centralized (or connected) facility or cybercollaboratory. The data system is very similar to the CUAHSI cybercollaboratory since Michael Piasecki is an investigator on the CUAHSI and LTER-BES projects. The major CI indicated in the proposal was related to integrated data management and handling. The LTER Baltimore Ecosystem Study (BES) has been in operation for 6 years, but data management and integration has been a serious problem. CI specific needs for the network involve 1) design of the network (i.e., data-centric versus federated system), and 2) level of data use or functionality that is desired by users who would access the network. A data-centric system is one in which all data are transmitted from sites to a central location for QA/QC, curation and archiving. A federated system is one in which all the data are held by individual nodes or sites within the system who then deposit copies of their data into a central repository; control of and responsibility for these data remain with the generator of the data at the node/site.

These needs were identified through review of existing data and survey of the data users. The project is envisioned as a value added effort that would expand the usability of data and knowledge generated as part of the LTER-BES project. CI development was described as central

for advancing progress in environmental informatics and decision-making for LTERs, specifically the LTER-BES. Major completed tasks include an analysis of how the individual investigators within the LTER-BES are collecting and handling data using the project's distributed data management system. Here, it was found that spatial and temporal data were stored in a variety of formats and at several locations (a federated system). A thematic categorization of data holdings, access to metadata, display of data access policies (network and site specific), project publications, and facts and information relating to the LTER-BES project and the investigative team is collected on the Open Research System (ORS) repository for the project. This prototype end-to-end system for data collection, storage, analysis and visualization is the primary deliverable produced from this planning grant. The greatest challenge noted was attempting to reshape the original distributed data management system to one integrated structure that can be configured as a node for the LTER-BES project with internal interoperability as well as external interoperability with other networks. The data collected for the past 6 years at this LTER will require a significant investment of resources to recast into metadata sets that can be incorporated into models and used widely for effective environmental system analysis. The most important lesson learned is to plan earlier for inter-operability and widespread data access.

Specific recommendations relevant to the WATERS Network planning process were adopt and enforce a centrally controlled standard for data publication (format and metadata); adopt a single design for an observational database, data model and description model; adopt consistent QA/QC policies, access control, software use; promote programmatic access for a wide audience (e.g., via web-services); develop detailed documentation on technologies used, user guides, examples, tutorials; maintain a flexible design framework such that continued advancements and upgrades can be incorporated into the information system.

Plan for the Blackstone River EFF and the Next Generation Cyberinfrastructure [Reckhow, D., Hanson, A., Long, S., Ganesan, D., Rees, P., Schultz, H., Kurose, J., Riseman, E., Osterweil, L., Dorner, S., Schweik, D.; University of Massachusetts at Amherst]

CI was a major part of the proposed project. One of the two technical objectives was to "formulate a general plan for the development of a "Collaborative Large-Scale Engineering Analysis Network for Environmental Research." CI was defined as including integration of databases, adaptive monitoring and control systems, sensor nodes, resource managers, user agents, networking, system and data management and visualization, and environmental modeling. The initial CI developed was considered critical to minimize inefficiencies that have hampered environmental research and to avoid unproductive repetition of experimentation and data collection. A lesson learned early was that domain scientists and computer scientists needed significant time and effort to learn each other's language and to find questions that engaged both communities. CI has not yet been developed on this project, but planning continues. The initial focus is on sensors and sensor development and integration. During this development several additional CI needs were identified that were not previously anticipated. These include nodes/proxies, data compression models, data rate management, matching data from disparate sensors, and uncertainty and data fusion. These final two were identified as of particular interest to the computer scientists.

A Large-Scale Environmental Research and Analysis Network of the Upper Mississippi River Basin (UMRB) for Hydrology, Sediments, Nutrients, and Emerging Chemical Contaminants [Schnoor, J., Brus, C., Carmichael, G., Hornbuckle, K., Just, C., Krajewski, W., Kruger, A., Muste, M., Papanicolaou, T., Stanier, C., Campbell, J.; University of Iowa]

CI was defined as databases, platforms, broadband, and high performance computing, and was one of five foci in this proposal. A major project goal was to evaluate the current CI and identify future needs in CI. CI was described as important because of the size of datasets and the known problems of managing large datasets, including archiving and searching. This project also discussed the issue of how computer scientists could assist domain engineers and scientists in producing seamless connections between different models, having different variables, different grids, different timescales, and possibly even residing at different locations. This project resulted in a digital watershed for Clear Creek, deployment of sensors, and two new web sites. The project web site is http://atmos.cgrer.uiowa.edu/WATERS/ .

Collaborative Research: Collaborative Large-Scale Engineering Analysis Network for Environmental Research for the Coastal Margin [Bonner, T., Jones, K., Wentling, T., Zaslavsky, I., Love, N; Texas A&M University, Howard University, UIUC, UCSD, Virginia Tech]

The ultimate goal of this project was to facilitate a cyberinfrastructure linked research network for national and international coastal margins research and policy-making. The envisioned CI would include a data repository, mechanisms for data aggregation, and computational tools for advanced modeling and simulation of complex coastal margins processes. The initial goal was to develop a data grid as an open standards-based system where heterogeneous distributed data sets, including real-time sensor databases, could be accessed and queried via a collection of middleware grid services that support information integration (WATERS Grid). The planning grant funded a series of five workshops to identify the current CI and how to integrate existing resources. The results of the planning workshops identified metadata, chain of control, and QA/QC issues as the most important related to CI. Secondarily, temporal/spatial resolution of data and the necessity of converters for existing data so that it can be “read” by models were identified as key CI concerns.

MEASURES: A Proof of Concept Demonstration [Multiscale Environmental Analysis in Urban and Regional Ecosystems] [Fernando, H., Anderson, J., Crittenden, J., Panchanathan, S. Arizona State University]

CI was identified as 1) data acquisition, compression and transmission; 2) computation and analysis; and 3) storage and visualization. The primary focus on CI was expressed in terms of two comprehensive high time-resolution monitoring sites. A proof of concept CI test bed was proposed to deal with data from these sites. The CI was presented as an essential component of the project in order to answer the science questions.

The project developed and implemented a skeleton network, which is being used for data gathering and storage and to run models. A new data mining technique was developed to predict atmospheric events. The CI is user-directed, in that the user controls data formatting and data visualization. Several critical CI development needs were identified in the work: the use of high performance computing to help design the systems, place sensors in optimal locations and enhance interactions between components and networks; sensor networking methods to achieve

most effective measurement, analysis, and prediction capabilities; data compression algorithms to allow rapid transfer and processing of large data sets; embedded sensor technology to allow autonomous or near autonomous operation of sensors on-board; QA/QC and other pre-processing needs (scalable, flexible, and robust networks that make use of Met (Java) Spaces and "browser like" methods such as web crawlers for handling ad-hoc networks where nodes may be static or mobile); development of small, light weight, low power and low cost sensors methods to deal with sensor vandalism; cyber security techniques for data assimilation into modeling tools within cyberinfrastructure; affordable mobile profiling and sensing platforms, sophisticated interactive tools (GUI etc), and protocols for operator versus artificial-intelligence intervention when needed; high-end sensor-model interaction (data-model fusion) techniques, wherein model output is used to rapidly and accurately adjust sensor placement to obtain best possible information power needs.

Planning a Multiscale Sensor Network to Observe, Forecast and Manage [Harmon, T., Bales, R., Estrin, D., Kaiser, W., Traina, S.; University of California – Merced]

CI was defined as supporting a distributed sensor network targeting the greater San Joaquin River basin. Sensor network and subnets were the defined CI, with software associated with managing these sensors also considered. A project website was established for interactions amongst the project team and external collaborators (https://ucmeng.net/people/tharmon/Current_Research/WATERS_project/). Four components made up the multiscale sensor network: (1) existing geospatial data for the area were identified, (2) human-actuated sensors were deployed, (3) stationary sensors were embedded, and (4) the robotic Networked Infomechanical System (NIMS) was used for high resolution scanning.

Recommendations from this project include:

- Funding several end-to-end WATERS test beds in order to put the planning outcomes to the test prior to large scale implementation,
- Stimulating collaborative industry-academic research and development efforts aimed at addressing sensor needs, and
- Focusing effort on engaging government agencies and watershed NGOs in an effort to integrate their needs into the planning process as early as possible.

Options for Field Facilities and Cyber Infrastructure in America's Heartland [Hondzo, M., Brezonik, P., Shekhar, S., Novak, P., Hozalski, R., Arnold, W.; University of Minnesota]

CI was the focus of this proposal which had the following objectives: (1) establish a pilot collaboratory that might consist of communication tools for the principal investigators; (2) elaborate on embedded network sensing, network topography, time synchronization, data routing, channel connectivity, and data storage/mining; and, (3) evaluate information needs for environmental field facilities and for the integration of EFFs with the national network (WATERS). Initial CI needs were described as computing, storage, database management and data transfer. The preliminary results have been related to deployment of a small network of sensors in Minnesota streams. No CI deployment has been accomplished and no lessons learned are discussed at this stage. One challenge identified was the use of domain students

(environmental engineers) for sensor and CI deployment. The major problem noted was that off the shelf data sensors were not all that easy to use and development of one new sensor (nitrate) was hampered by lack of ability of the environmental engineering graduate student in IT aspects. The high level of skill necessary to use the prototype level CI was unanticipated.

A National Infrastructure for Environmental Analysis Research and Education [Pancake, C., Hanus, F., Ferschweiler, K., Haerer, S.; Oregon State University]

This proposal was focused on CI with a goal of evaluating IT infrastructure and coordination infrastructure to make WATERS a "collaboratory." Several critical CI aspects were recognized as important including: field facilities with streaming data capabilities, shared repositories of data, software tools to support user tasks, collaboration environments, and web-based portals to support user tasks. Usability was seen as a critical aspect. It is important that the CI identify and respond to user needs otherwise the infrastructure will be underused. Flexibility, extensibility and scalability were also highlighted. Flexibility was considered critical because the community is heterogeneous and the roll out will take place over time. Extensibility and scalability were considered to be important because of expected advances in these critical areas during the implementation phase.

Nine communities were reviewed on this project: CUREE, EERI, LTER, NCEAS, NEES, NEON, NPACI, ORION, and UCAR. Best practices were identified and summarized in the final report (not available yet).

Existing commercial and open source products for CI were evaluated – 87 products overall (also detailed in the final report).

User requirements identified included: executive collaboration, committee collaboration, research collaboration, collaborative calendar and scheduling, and workshop collaborations. Products evaluated fell into several categories: virtual discussion products, virtual meeting products, virtual seminar products, document sharing and management products, application sharing products, and web-based organization products. The review involved three iterations. First, constraints or limitations were noted that would make the product unsuitable for the WATERS community. Second, the products were characterized in terms of functionalities, operating systems, and number of users. Finally, products with similar functionalities were compared with each other. Fifteen products were ranked highly in evaluation of usability and features. Detailed comparisons were developed for these fifteen. It was noted that three types of products were available for the desired uses: hosted services, software products and service/software server options.

WATERS: Cyberinfrastructure and Management System Development for the National Collaborative Large-Scale Engineering Analysis Network for Environmental Research Network [Eheart, J.W., Welge, M., Minsker, B.; University of Illinois at Urbana-Champaign]

CI was defined in this project as computational infrastructure, data storage and management, data sharing, data fusion, assimilation, visualization, and mining, model sharing and archiving, knowledge management, decision support, and uncertainty accommodation. The focus of this project was on developing a consensus of needs in CI for the community (very similar to the

goals of the CI Committee Report). Further, the project planned to define the CI needed to support the WATERS Network including needs that can be met with existing IT and those that require development of new technologies. The cybercollaboratory being used by the WATERS Planning grant was developed as part of this project (http://WATERS.ncsa.uiuc.edu/cybercollab). There was no discussion of why this CI development was first, but it was indicated that the cybercollaboratory would assist with the evaluation of other CI needs. Use-case scenarios were developed using an interactive WIKI to help the community understand how CI might work in their own areas. A survey report was created (Bhagwat, 2005). Further, the process of deploying a testbed cyberinfrastructure led to identification of critical issues.

Developing user-friendly CI requires a tremendous investment of time and resources. Particularly essential is the commitment of community members who are willing to invest significant time in working with developers to ensure that the software meets their needs. Having tolerant, motivated early adopters who need the software to accomplish their research, service, or educational goals is essential. The early adopters should be funded to work with the developers in a fully synergistic development effort. An iterative development process is needed, where the developers observe the community members, discuss their needs with them, and create prototypes; community members actively use the prototypes for their research, service, or education and provide feedback; formal usability studies are conducted where community members are asked to perform tasks with the prototypes while researchers observe their responses; developers modify the software to address the concerns raised; and the process repeats. Community liaisons need to be designated who have excellent communication skills and can assist the early adopters in learning how to use the prototypes, to avoid the frustration that can come from learning to use new technologies, particularly ones that are unproven and not yet well documented. CI development efforts need to have sufficient resources to allow for this intensive process, or the CI will likely have poor acceptance by the community.

WATERS: Collaborative Research: Developing a Plan for a WATERS Neuse River Basin in North Carolina [Pierce, J., Reckhow, K., Prudhomme, T.; Duke University and University of Illinois at Urbana-Champaign]

CI's role, as defined in the proposal, is to facilitate the "linking of data collection networks (sensors) and forecasting models for adaptive implementation of nitrogen management plans in the Neuse River basin", with specific focus on methods for combining disparate information. CI components include "broadband networking infrastructure, advanced computational systems, data storage facilities, data and software repository technologies, and middleware (grid) technologies that are currently available to science and engineering research communities." "Linking information technology to research scenarios" was also identified as the key step in defining cyberinfrastructure.

The CI development focused on data collection, data repository creation and maintenance, data access and modeling efforts within Neuse River Basin. The main component is organizing a local network of scientists and engineers from five research universities in the area. This local network of people will then lead the Neuse River WATERS Facility, which will make Neuse River data repositories at all universities accessible to all partners and to support various modeling efforts. Another component of CI development is integration with the CUAHSI effort,

where Neuse River watershed is used as one of "digital watershed" prototypes. Particular CI recommendations of the project are still a work in progress.

WATERS: Collaborative Research: Concept Development Toward a Collaborative Large-Scale Engineering Analysis Network for Environmental Research with Focus on Chesapeake Bay [Ball, W., Amir, Y., Burns, R.; Johns Hopkins University]

CI is defined as a framework for fusing the growing body of scientific data with computational models, to describe, analyze and forecast the physical, chemical and biological properties and processes in the Chesapeake Bay. In this area, intense relationships between human activity, water quality impairment, and vitality of living resources require specific management approaches. Multiple though fragmented monitoring and modeling programs co-exist (i.e. new sensor deployment is not critical for prototyping CI), and hence cross-domain integration is necessary, in the form of CBEO (Chesapeake Bay Environmental Observatory) as a local CI prototype.

The important components of the CI model in this project are: a) focus on a clearly defined cross-domain scientific problem (hypoxia in coastal waters), in which hypotheses can be evaluated through integrated data analysis and modeling; b) availability of monitoring systems that can be integrated to address the hypoxia problem; c) data integration challenges, including standards-based data publication, description and discovery, resolving semantic and structural heterogeneities, spatiotemporal interpolation, managing different sampling methods, and assembling integrated datasets for modeling and visualization; d) management of model-derived data; and, e) reliance on existing cyberinfrastructure efforts (GEON node, in particular), to share data, research tools and models.

Development of Informatics Infrastructure for the Hydrologic Sciences (CUAHSI HIS) [Maidment, D., Baru, C., Piasecki, M., Kumar, P.]

The CUAHSI community has been considering the meaning of CI for several years. CUAHSI recently released a report on their activities, *Hydrologic Information System Status Report*. (CUAHSI 2005) This report focuses heavily on the development of the "Hydrologic Information System – a combination of hydrologic data, tools and simulation models that supports hydrologic science, education, and practice." Critical cyberinfrastructure for this community is focused on data access and visualization. Unlike other large data-intensive scientific community, hydrologists use "third party," often governmental data more intensively. These data sources are usually not interoperable, thus a significant need is for cyberinfrastructure to allow seamless access to multiple data repositories. For this and other reasons, CUAHSI has elected a portal structure as a mainstay of its CI. A portal[15] is "a structured computer interface environment that integrates many kinds of information products and services from disparate sources." Further, CUAHSI has conceptualized a "digital hydrologic observatory" focused on natural boundaries (watersheds) as a framework for data storage as well as visualization.

[15] Portal is a generic term for the system that integrates disparate sources. A web portal is specifically focused on integrating sources from the internet. CUAHSI is currently focused specifically on web portals, but expects the more general portal structure to outlive reliance on internet resources alone.

Appendix E. Cross-cutting Concerns that Affect Design Choice

Interoperability

The WATERS Network will incorporate resources managed by other projects and agencies, co-exist with a range of other environmental observatories, and live in the context of the broad development of CI and CI-enabled science and engineering. Interoperability and re-use across this ecosystem is clearly required to enable the WATERS Network science plan as well as to build and operate the Network in a cost-effective manner.

The most obvious approach to achieving these goals would be to have all participants use the same data model, the same metadata standard, the same observations database design, the same modeling frameworks, and the same service architecture, and to manage change through globally coordinated updates. However, upon analysis, it becomes clear that the coordination and opportunity costs of such a monoculture CI design would outweigh the benefits. The Environmental Observatory Systems (EOS) individually and collectively spans historically independent communities that, through a combination of tradition, preference, and real differences in requirements, have made different information technology choices. While the national-scale, systems-level investigations driving the creation of the EOS motivate closer coordination, they do not eliminate the domain-specific investigations, the independence of funding sources, and the benefits of being able to innovate independently that have led to the current situation and will continue to work against large-scale coordination.

It is clear therefore that, while standardization may increase, the WATERS Network CI as a whole must be designed to manage continuing heterogeneity. At a design level, this implies the WATERS CI cannot, for example, assume uniform, static data types and formats that can be hardcoded into databases and software components. Instead, the CI must explicitly manage heterogeneous resources by type and should support translation/bridging between them. For this to be more than a simple laissez-faire approach, the CI should also provide recommended/default options, foster dissemination and adoption of best practices, and enable collaboration around standardization efforts.

Quality control assessment

Successful implementation of cyberinfrastructure (CI) for the WATERS Network will require rigorous validation and testing of applications and services. Fortunately, there are good models for quality control and assessment from prior CI projects. Specifically, NEESgrid – the system integration component of the George E. Brown, Jr. Network for Earthquake Engineering Simulation (2001-2004) – pioneered the use of acceptance testing for the various systems that made up the NEES collaboratory. The NEESgrid legacy web site contains details about the testing program (http://neesgrid.org/acceptance/). In addition to the tests described below, NSF also hired an outside firm to take the NEESgrid documentation and installation, attempt to build the system, and then report on the ease or difficulty of this effort.

Within NEESgrid, the acceptance testing program took three forms. First, the project used a set of protocols, termed “component unit” tests, to confirm that individual software components were functionally correct. Specifically, developers produced a unit test document that specified the operating environment for a particular application or code, a set of test commands or entries made using the code, and the expected results that would be produced assuming successful operation of the code. In the NEESgrid project these unit tests would typically be performed by

the developers and observed by representatives of the customer (in this case, the NEES Consortium in the form of members of the NEES IT committee). Once completed, the results of the test were documented and stored. In the event of unsuccessful tests, steps were taken to amend the malfunctioning code and fixes were verified in a subsequent unit test – with this cycle repeated until the code could pass the test.

The second element of the NEESgrid acceptance testing program consisted of "workflow" tests, or end-to-end operation of linked systems. For example, a typical workflow test in the NEESgrid case involved acquisition of data, storage of acquired data, and subsequent retrieval and visualization of the data. Unlike the unit tests, which largely focused on success or failure, the workflow tests also examined performance, such as whether a particular set of actions could be completed quickly or whether a particular service operated under heavy load (e.g., simulation of hundreds of simultaneous users). As with the unit tests, workflow tests were conducted by the developers and witnessed by members of the NEES IT committee. Once completed, tests results were documented and stored. Failed tests were re-run until the workflow test succeeded. Finally, there was a third element of the testing program that focused on packaging and installation. That is, could a third party download and install the NEESgrid software components.

The NEESgrid approach offered a number of advantages. First, it required system developers to articulate specific success and performance criteria. Second, it created specific milestones (e.g., the scheduled tests dates) that helped discipline software developers. Third, it provided a way for the customer, in this case the NEES Consortium, to certify successful operation of applications and services before accepting delivery. Despite these strengths, there were some liabilities associated with the testing program in NEESgrid. First, the testing effort was time consuming and added an unaccustomed level of bureaucracy for university-based software developers. Second, the testing arrangement created an adversarial situation between the code developers (NEESgrid project team) and the customer (NEES Consortium) which sometimes undermined effective communication. Third, the tests created the illusion that successful completion meant the system was usable – when the tests actually defined a minimum level of acceptance. A more rigorous testing program would have involved subjects attempting to use the developed systems (i.e., the way software is tested by Microsoft and etc.), where this kind of usability feedback is often more critical in shaping evolution of applications.

Scalability

When building a permanent facility, a critical element of planning is to design its infrastructure to meet not only current needs, but the anticipated needs over the full lifetime of the project. Traditionally, in sizing a facility building for example, this involves initial over-provisioning. With cyberinfrastructure for observatories, this can be an extremely costly approach due to the combination of exponentially decreasing costs and exponentially increasing demands. Most people are familiar with Moore's Law or some variant that capture the trend that processing power per processor or per dollar has been doubling roughly every two years. Storage, network bandwidth, and sensor capabilities are also doubling with periods much shorter than the lifetime of an observatory. These trends affect not only costs but have a further feedback on demand; increased capabilities make it possible to address additional research issues and/or to adopt more resource-intensive experimental designs. Thus, in designing CI, it is important to adopt scalable designs with just-in-time provisioning instead of initially scaled (and potentially less extensible) designs.

While a detailed technical architecture for the WATERS CI cannot be specified without further investigation and additional detail on the specifics of the science plan and other aspects of WATERS Network, the general scope and cross-cutting concerns described above do constrain the design space sufficiently to identify an appropriate set of design principles and overall conceptual architecture. At this level of specification, the WATERS CI should be an anticipated-best-practice design for a scalable, distributed, multi-objective CI for long-term scientific knowledge creation and dissemination as part of a larger knowledge ecosystem. Framed this way, it is clear that WATERS can draw from CI design efforts from many disciplines, particularly those that address issues of long-term persistence, evolvability, and interoperability, and conversely, that the discussion and design information gathered here should be relevant for other environmental observatories and next-generation community cyberenvironments in general.

At the most fundamental level, the WATERS CI must be architected for change: change in scale, change in component location, change in data types and community processes, change in scientific understanding, change in technologies, change in the community, and change in the overall scientific ecosystem. The most appropriate design response, again at a fundamental level, is to incorporate capabilities for managing change in the system and to decouple the most rapidly changing aspects from core capabilities and from each other. For example, if resource location can change, the system should use global identifiers that are independent of location and provide a resolution mechanism to provide a current location and access method(s) for a resource given its identifier – the basic idea of address books and phone directories. The following subsections continue this type of mapping from requirement to relevant design principles.

Distributed Content Management

Content management (CM) recognizes that, independent of the type of resource (content), we have common needs to store, locate, version, describe, relate, control access to, annotate, and convert them. The content being managed could be data, documents, software, workflows, services, sensors, compute resources, storage resources, etc. With a CM abstraction, the types of content being managed, how that content is described, and even what processes are applied to that content can be changed without modifying the underlying software; such decisions are managed by the end-user (or by administrators on their behalf) rather than by the software developers. To enable CM across a distributed system then requires the use of globally unique identifiers for content, content types and descriptors and resolution/look-up mechanisms as introduced above. Various aspects of CM can be seen in web content management systems, laboratory information management systems (LIMS), grid metadata and replica catalogs, document and data sharing portals, semantic grid/e-Science systems, etc. In the context of WATERS Network requirements, CM as a design principle is an effective means to support the continuing evolution and extension of the types of information managed by the system and to decouple this from the specific initial and upgrade choices made for underlying storage technologies.

Distributed Process Management

Somewhat analogously, designing in terms of process management (PM) recognizes that, independent of the details of individual processes, we have a need to create and share them and manage their execution. A PM abstraction involves capabilities for creating explicit descriptions of processes at multiple layers of abstraction (e.g. scientific, mathematical, service-level

template, service-level instance), the ability to semi-automatically move between layers, and the ability to execute low-level descriptions with minimal human intervention. Workflow systems, which have traditionally focused only on the last step, provide a key starting point but a wider infrastructure to manage process descriptions (e.g. as content that can be shared and versioned), execute descriptions on an evolving mix of local and remote resources, connect processes that cross organizational boundaries, and automatically track provenance, is needed. By defining processes in terms of clearly defined steps and by separating the specification of processes from the development of software to execute them, processes can be dynamically defined and evolved independent of the choice of whether to run them on community or third-party resources and independent of the evolution of the underlying hardware and software systems.

Virtual Organization-Based Management

The concept of virtual organizations (VO) recognizes that distributed groups have requirements analogous to those of traditional organizations to manage membership, policies, resources, procedures, organizational knowledge, etc. VO-based management then recognizes that, independent of the specific choices made by a VO, its membership and the software and hardware systems composing its CI have a need to align their behaviors with those choices and that explicit management of overall VO context lowers the barrier to incorporating new software and bringing new members up to speed. Some of the clearest examples of VO-based management in current CI are the management of authentication, authorization, and allocation policies in grid systems and the management of group context (.e.g. shared calendars, discussion lists, and document repositories) in collaborative portals. These concepts can be extended to the broader context of the WATERS Network CI to manage observatory-wide agreements about, for example, access policy, preferred/supported content types, required metadata, data pipeline configuration, and curation procedures. The same mechanisms could be used by groups in the community to manage their operations and the combination of resources allocated to them by WATERS and available from other sources (e.g. enabling them to trigger a model run based on Network sensor readings that will execute on a local cluster made available via a separate grant and to perform their own quality analysis before releasing the results to the larger WATERS community).

Provenance Management

Designing for research coordination with the scope, depth, and duration envisioned for WATERS necessitates the adoption of one additional design principle; given the complexity of the overall system and the expectation of continuing change, provenance – the broad historical context of data and resources – must be managed explicitly. In some sense this is simply a recognition that relying on tacit knowledge of how data was created and on off-line sources such as paper notebooks does not scale to the observatory level. At a practical level, this implies an infrastructure to document the version of all resources and the detailed processes they are used in that lead to the creation of data and new resources (e.g. to document the set of tests performed on a new analysis routine). While the previous design principles contribute to the development of such a capability, it is important to emphasize in the design that users will potentially require access to the provenance of a given resource, be able to compare its provenance with that of similar resources, and be able to add to that provenance, at any time they access the resource. This principle also implies that in addition to tracking the provenance of WATERS observations themselves, the CI should be capable of managing the provenance of community research (as a

community service to maintain the link to WATERS data and to maintain a full history of results that may affect subsequent changes in WATERS itself).). This capability is important to enable community members to identify related resources to the data they are accessing (e.g., publications and analyses), but also for understanding what quality control procedures have been applied to the data.

Open Interfaces and Protocols

Another design principle, somewhat a corollary of the previous ones, is the use of open, standard programming interfaces and protocols as a means to assure end-to-end functionality without constraining implementation. Further, this principle favors the use of limited but broadly adopted standards in preference to more comprehensive ones with limited implementation; discovery mechanisms and additional management interfaces can be used to expose functionality beyond the standard interface. This approach helps avoid lock-in to a particular commercial or research implementation path. For example, the Java Content Repository (JCR) standard covers a wide range of capabilities for creating and managing resources and their properties but does not specify the means of managing content types. This has led to a range of commercial and open source implementations and, while it has not eliminated all dependencies on the particular implantation chosen, it has significantly improved the interoperability of JCR-based applications. Using this approach across the WATERS CI is expected to maximize the choices for internal implementation as well as the potential for direct interoperability with other CIs.

Appendix F. Architectural Components for WATERS CI

Introduction

As noted throughout this document, the WATERS Network will exist as part of a larger cyberinfrastructure ecosystem and will need to leverage and interoperate with infrastructure developed for other observatories and for other scientific domains. The following sections discuss technology areas, moving from core infrastructure to the user level environment, highlighting specific components and capabilities that will be needed within the WATERS Network. Within this discussion, areas where the WATERS Network should be able to leverage existing technologies and areas where the Network may represent a driving application are noted.

Base Technology: Computation, Storage, Communication

As with all cyberinfrastructure, the WATERS Network will ultimately rest on computation, storage, and network technologies. While the aggregate needs in these areas are significant and beyond the scale previously available in this discipline, they are none-the-less likely to be modest initially compared with communities now driving towards the petascale. Network capacity will be required for real-time streaming of WATERS observations as well as delivery of real-time and historical aggregates of observations, derived and modeled data, and other digital artifacts (e.g. documents) to users. Computational resources will be required to perform near real-time QA/QC and generation of derived data products as well as to run analysis workflows and simulations, integrate and mine data, generate visualizations, etc. Storage will be required for raw data, metadata, derived products, and other data and documents generated by the community. In all of these areas, relatively straight forward calculations based on number of sensors and data rates, computational complexity of models, and predicted user demands will enable rough estimates of the overall capacity that will be required. Depending on the final configuration of the Network, these requirements could range from relatively modest, i.e. requiring no more than a small cluster, to significant, i.e. requiring traditional or on-demand high performance computing resources, networks, and storage facilities. For example, if the sensor suite includes a significant number of video feeds that require processing, i.e. the use of feature extraction algorithms to identify organisms or transient phenomena, or if there is significant processing required to incorporate third-party remote sensing data, the real-time processing needs would be dramatically larger than for a suite of point sensors. Similarly, user requests for historical data, or to create custom data products on demand, to run high-fidelity models for comparison with observation, or to directly apply modeling to optimize sensor positioning and overall experimental design or perform research (e.g., for predicting the performance of potential management practices) would require much more significant infrastructure than simply serving recent data.

To a large extent, the WATERS Network should be able to leverage commodity resources and capabilities at this level, with the potential exception of requirements related to on-demand/stream processing capabilities that, while potentially common across observatories, is not yet well developed. Once the scope of requirements is set, it will be possible to scope the core infrastructure needs for the WATERS Network in more detail and to identify potential trade-offs between computation, network bandwidth, and storage to minimize costs (i.e. indexing and caching increase storage needs but reduce the bandwidth and on-demand computational needs, regenerating derived data from stored workflows rather than storing the originals has the

opposite effect). Together, these kinds of analyses will enable specification of a suite of resources that may involve shared resources such as the TeraGrid as well as dedicated, on-demand availability of computation, network bandwidth, and storage resources owned or provided as services to the WATERS Network.

Operating Systems, Middleware, and Services

Above the layer of raw computational power, the WATERS Network will rely on a layered set of software systems including operating systems, middleware, and services. These range from the more basic tools such as operating systems, cluster management tools, schedulers, and cybersecurity capabilities to higher level tools providing capabilities in areas such as data management and integration, observatory management.

The current infrastructure in science and engineering domains to be served by the WATERS Network is very heterogeneous and interoperation is difficult at best. Moving data across community boundaries, translating it, preparing it, and ingesting it to the next application is hampered because the mediation services do not yet exist to the extent necessary. For example, it is still a major undertaking to try to duplicate certain model runs done in a specific region of interest and attempting to use another numerical model to repeat these runs even if all initial and boundary conditions remain the same.

Solving this problem will require a combination of adopting the design principles outlined in Appendix E and adopting standards for data formats and models, as well as middleware programming interfaces and protocols as much as possible. Given the fact that legacy formats, applications, and storage are not likely to change dramatically (the previous investment of manpower and finances prohibit this), interoperability will likely involve wrapping legacy software and data structures to map them to community standards (a hub and spoke integration model) which other communities and CI developers can operate and program against. If all of the EOS can do this, then any other EOS can take this interface and make use of it based on its own community needs. This means that interoperability will be placed in an intermediate layer that will largely deal with mediation services.

In this regard, it is crucial that the WATERS Network adopt an open service architecture that exposes the data holdings and applications used via programmable interfaces and protocols are widely adopted in industry and able to run on commonly available hardware and interface with common software such as browsers and applications such as EXCEL or MATLAB. CUAHSI is one of many examples where such an approach based on use of web-services (in this case focusing on the many different tasks associated with data search and query, data access, data ingestion, and data translation) is being explored with positive results. The WATERS Network will clearly need to adopt industry standards at this layer of infrastructure which would significantly ease interoperability issues with other observatories and third-party data and software providers taking a similar approach and most directly, would allow integration with the services currently published by CUAHSI.

It should be noted that there is a relatively steep learning curve to understand web service technologies (such as SOAP or WSDL) and that true usability and interoperability will require the WATERS Network to document these services, provide examples, deliver some basic plug-in

applications and macros, and expend effort and energy to train users on the use of these technologies.

In the next three subsections, the potential for standardization related to data formats, metadata, and logical data models are explored.

2. Data Models and Formats

To enable data sharing, the WATERS Network will require a combination of standards and translation capabilities, at the level of logical models as well as for physical file and database formats.

A common data model that can be used as the underpinning structure for continuous or discrete data in space and time is a key component for CI development. This is particularly important in view of the many variables that the environmental engineering community is interested in. Because of the cross EOS interest in data holdings of other EOS', interoperability in the data model world is crucial so representations in space and time in one community are sensible to others and can be transferred between the various IT systems.

The CUAHSI community has used a data cube model (three axes that depict variable, time and space) approach that permits the unique identification of any data item (see the status report on the CUAHSI HIS efforts). Such a model is also implicit in many of the file formats in use today. If the WATERS Network does not fully standardize how this observational model is represented then, for reasons of interoperability, it is important that the structure and access to it be documented and made available through an interface that allows programmatic access and interpretation, i.e. as part of an open service architecture. The publication and exposure of services that permit access to WATERS data holdings via well defined interfaces that can be included in common end user application is crucial for interoperability among all EOS and should be considered a must.

The same arguments brought forward in the previous paragraph apply to a data base design for data collected as part of the WATERS efforts, i.e. either new data on the site or data that has already been collected and is slated to be ingested into the WATERS data base. CUAHSI has been working on the design of a Hydrologic Observations Database (HOD) that is targeted at ingesting the (mostly) time series data collected in the network. Adoption and subsequent expansion of the already existing design would be, as mentioned before, a benefit to the WATERS Network CI development efforts. Regardless of the choice made, interoperability across the EOS and with data sources from other agencies will require that the design of the database be registered, documented, and that access is made available through well designed and documented automatic services.

A common data format that fits all is likely to be unachievable goal even in the long run (less likely than agreement on logical models). The fact is that many communities have either agreed on one or two standards (like seismology where the SEGY and SEED formats are very popular) or other communities (like CUAHSI and WATERS) where a plethora of formats, both standardized and personal creations abound. These adoptions are typically driven by historical

reasons ("I have always used this format starting from my student days") or by popular analysis and modeling applications. Unity in this area of CI will be impossible (the coordination costs will outweigh the benefit) which mandates the need for accommodation and translation.

One example is the OpenDAP system that exposes a layer of format translation services so that data in one format can be downloaded as another format. These kinds of services are crucial for broad interoperability and the WATERS Network will need such middleware available, for direct use by researchers as well as embedded as part of other CI development. The hydrologic science research community routinely deals with application dependent formats, for example for simulation models like HEC-RAS or FLDWAVE, but also need to ingest data formats used by large mission agencies like netCDF, HDF5 or GRIB.

The degree of interoperability would be increased if the WATERS Network could agree to adopt one of the widely used flexible data formats like netCDF or HDF5 as a default. These formats are powerful and widely used and have libraries associated with them that deal with en- and decoding, which means that many end users have utility programs available to ingest them. If the WATERS Network decides to create new formats (or profiles within the netCDF/HDF5 frameworks), documentation, access libraries, and web-service translators will be an essential part of promoting interoperability. In this context the WATERS Network would be able to foster and promote interoperability by actively developing translation services and making those available to all EOS.

3. Metadata

Metadata descriptions will play a crucial role in making interoperability a reality. Metadata will not only be used to describe the semantic and syntactic aspects of data collected or generated, but will also be used to describe services, data sources, and even workflow sequences. As such, metadata will have to be rich in both quantity and quality in order to accommodate all the needs and end use scenarios. On the other hand, metadata constitutes a considerable extra burden as far data generation is concerned and always runs the danger of being perceived as unnecessary ballast. Obviously a compromise must be found, probably involving significant automation of metadata collection that will ensure that as much as is needed is in fact available.

In view of the statements made earlier, metadata descriptions will vary as one moves across the communities. Hence, it is important to stress that one cannot attempt to force any of the EOS to use just one description standard. While a healthy discussion on what works best is certainly a desirable activity, it is inevitable that the EOS will use different descriptions. For example, NEON is slated to use the Ecologic Metadata Language (EML) because it is a description framework that works for ecologists. ORION on the other side will be more inclined to work with the IOOS Data Management and Communications (DMAC) document, a selection that makes sense as it is rooted in the ocean sciences already. On the other side the CUAHSI branch of the WATERS Network is using the ISO 19xxx group of standards for their description, while access to governmental data sources are likely to expose a description framework that is based on the Federal Geographic Data Committee (FGDC) recommended content standard.

It is clear from the above that there will be no ONE standard, actually not even a comparable profile even if one of the EOS would use the same standard. Again, this must be accepted as a fact and one should not try to even approach the interoperability need from the point of attempting to force a common framework on everyone. Rather, the key for metadata interoperability will be to develop a layer of mapping services that will mediate between the different profiles and the vocabulary used in those. Interoperability must mean that an environmental engineer in the WATERS Network is able to find a data set that has been collected (and is stored) in the ORION information system. It is not necessary that the ORION system describes it such that it also makes “sense” to an environmental engineer. What is necessary is that the mediation services make it “appear” to the environmental engineer as a sensible data source.

In view of the WATERS Network acting as an umbrella for both the environmental engineering and hydrologic science communities, it would be beneficial if the WATERS community would adopt the ISO 19xxx standard family for metadata development, and even use the CUAHSI community profile as a base to build their own profile. This also includes the use of ontologies for describing the controlled vocabulary, which is essential when attempting to unify semantic heterogeneities. It is crucial that the metadata profile is implemented using a machine readable schema, like XML schema or RDF. It is also important that the structure and relationships of the metadata can be parsed automatically so another community can learn about and ingest it. Vehicles like XML, RDF, OWL, and the Semantic WEB are indispensable technologies that greatly aid in achieving interoperability in addition to a comprehensive and sound documentation effort that clearly explains what is meant by what metadata tag, how it has been implemented, what the underlying rules are, where to find additional information, and the publishing of samples and tutorials. An extra benefit is that use of technologies like XML, RDF, and OWL provides the basis for mechanisms that allow curation and updating of metadata as well as inclusion of provenance metadata (e.g., assumptions behind calculated parameters used in a computational simulation). Adopting such standard tooling for managing metadata and ontologies, as well as leveraging metadata content standards in use within CUAHSI and other observatory-related efforts would significantly improve the interoperability and software reuse across the EOS.

Metadata interoperability is only possible however, if actual metadata is provided. While this may be obvious it is important to note that every metadata instantiation requires some work. It is well known from previous experiences (for example EML) that if the system is too cumbersome to use, then little metadata will be supplied. The key here is that as much metadata as possible should be generated automatically without manual data typing efforts. If this approach is maximized, then even extensive metadata sets can be annotated (this is really needed in terms of interoperability), without deterring a user community. This requires a “smart” metadata management system, a system that goes well beyond what is currently available, for example the MORPHO tool, an application used by the LTER community to annotate EML based metadata. This is an active area of development, within CUAHSI and in projects in other scientific domains, which should enable the WATERS Network to start with baseline capabilities significantly beyond what exists today.

Knowledge Management, Visualization and Modeling

Another major component of cyberinfrastructure involves data, its analysis and visualization. While these were separate areas in the original vision in the Atkins Report, the current NSF CI vision incorporates these into a single area. This is shown in Figure 2 as one of the critical pillars supporting the work in the collaboratory above.

Particularly within the environmental field, secondary data are extensively used by researchers. For example, many researchers access USGS or US EPA databases that include years of data on flow and chemistry in streams and rivers in the US. Maintenance of these databases is the responsibility of the sponsoring agencies and is out of the control of the researchers who depend on these data. Thus, a significant component of cyberinfrastructure is the knowledge management institutions and the information technology tools to enable adequate management and long term curation of critical datasets that already exist and those that are anticipated to become available in the future. Content management systems (CMS), digital libraries, and semantic technologies are likely to play a key role in providing the required capabilities in these areas and significant advances are likely through programs to support NSF's vision of a national data infrastructure that the WATERS Network will be able to draw from.

This knowledge management is critical for the already developed and continuously updated databases such as USGS and EPA's STORET, but even more important will be routine capture and management of raw data from all scientific endeavors. As discussed above, traditional publication includes limited or summary data while access to the original complete data set may be available in text format (generally as a PDF) at present. The goal of knowledge management will be to enable researchers to make their complete original data available in such a way that its long term archive is ensured and it can be easily and meaningfully accessed by others.

Integration of knowledge management with data analysis and visualization will allow researchers to use models to pursue research questions unaskable previously. For example, consider the case of zebra mussel movement along navigation rivers. The past two years represented the highest and lowest flow years on record for the Ohio River Valley. Not surprisingly, the zebra mussel population showed profound shifts in habitat and populations numbers. It would seem obvious that this was a flow-dominated event; however, without data on many other parameters affected by flow (e.g., concentration of nutrients, percent change in available habitat, change in water use and wastewater discharges by humans along the river, etc.) understanding how this observation might advance efforts to control this invasive species is limited. But consider an integrated database that contained all these data and allowed researchers to postulate relationships among these parameters, use these hypotheses to drive predictive models and compare these results with new observations (collected in near real time). The potential to provide critical insight to enable restoration of our waterways is significant.

In addition to access to data, researchers need access to information in a world seemingly overflowing in it. Knowledge services are those tools that connect people with information (like Google for general searches). These tools direct people to information in specific areas or connect people to the network of other people and places studying related topics.

Finally, a critical benefit of CI is the ability to re-use data within and across research domains. Re-use implies that data can be discovered, their semantics understood, and then applied – beyond the group or investigator that originally collected the data.

4. Modeling and Visualization

For it to be successful and embraced by the community, CI must enable academic and professional segments of the ENVE water environment community to evolve through a transformational process in which modeling and visualization are ubiquitous and easily accessed. This evolutionary process will rely heavily on human implementation and interaction with the following primary components: high performance computing-based infrastructure; intensive data management, access and sharing; data fusion and mining; model development and sharing for knowledge generation; and visualization for value enhancement and communication (Minkser ref.).

Despite the rapid growth in computational capability over the last two decades, the significant cost of modeling and visualization (MOD&VIZ) continues to limit its practical application. For the purposes of this discussion, the terms "practical application" and "traditional approach" refers to the routine use of MOD&VIZ tools within the broadly defined areas of environmental and water resources engineering and science. This occurs because the traditional MOD&VIZ practice is primarily PC-based, and most software, such as the widely used Excel, Matlab, and ArcGIS packages, are implemented in single processor machines. These modeling approaches fail to take advantage of high performance computing developments such as advanced computational strategies, high bandwidth communication through modern cyberinfrastructure, dynamic visualization tools, and sophisticated real-time model solving concepts. While there has been some research progress on individual elements of MOD&VIZ in a high performance computing environment, this progress tends to be highly focused and segmented, and has not yet found wide application in practical modeling and visualization efforts.

Model-based simulation and visualization within a high performance computing framework will create a new "window and vision" of the natural world. One that is rich with information, and able to convey complex interactions among, and within, complex systems that are impacted by human activities, all within a realistic setting. Such a framework can be defined as a very high order modeling process. Model-based simulation has been used extensively in the areas of atmospheric/weather, ocean dynamics, biomedical imaging, genomic research, and space science; leading to greater insights and understanding of fundamental "grand challenge" questions. Our understanding of environmental engineering and science related systems will benefit greatly from similar model-based research and intense visualization. For example, models have potential to address and solve fundamental water environment engineering challenges that include: surface water and groundwater interaction, field scale behavior of spatially variable (i.e., heterogeneous) processes, multi-scale processes that transcend spatial scales from nanoscale to global and time from micro-second to eons, connections between different aquifer systems and interplay between geologic, hydrogeologic, biogeochemical processes and human interventions. In addition, model-based control has the potential to merge sensor networks with models and feedback loops to enhance performance, reliability and economy of water supply, wastewater (Beck ref) and storm water infrastructure. Additional exploration and development that will rely on multidisciplinary CI and domain oriented research might focus on the following concepts:

- Warping grids, adaptive meshing and adaptive time scales to deal with multi-scale space/time challenges within a numerical modeling framework;

- Steering the simulation in which the modelers guide the process in real-time as an equal partner with the computer to realize greater insights into a given study area, and fundamental processes and relations that take place within the given area of interest;

- Exploring complex adaptive management concepts in which built infrastructure systems (i.e., water supply well fields) are able to adapt to natural and human induced stresses such as drought and/or disasters, and optimize their operations to be more efficient through predictive control involving both the model (i.e., automated) and active (i.e., human) input;

- Employing dynamic visualization of the modeling process in which interactive intervention is possible and encouraged, with heavy dependence on geographic information systems and translation through multiple space/time scales;

- Creating a "model-free" environment in which the modeling process is "turned upside down" and simple questions are asked of a knowledge kernel. Approaches to this concept might involve a hundred thousand model runs, combined with sophisticated data mining using soft computing techniques, and development of a knowledge kernel that captures the "essence" of the geographic area of interest.

Simulation modeling and visualization creates a unifying framework for synthesis of field information, improving sampling strategies, testing hypotheses, and identifying optimal management of complex systems that minimize cost (including environmental degradation) and maximize production. In fact, MOD&VIZ should play a key role in planning data collection strategies, data network designs, and sensor technologies to minimize the cost of data, and maximize the usefulness of the information. In addition, advanced modeling methods are able to consider imprecision and uncertainty in input data through Bayesian and related statistical approaches (Reckhow ref), and soft computing approaches such as fuzzy set theory (Woldt ref), thereby allowing the MOD&VIZ process to move forward, without having to wait until the perfect data set is available.

While research developments have provided a foundation for simulation-based modeling in a high performance computing environment, additional research and development will be required to bring a truly interactive modeling environment into a high performance computing framework. A different, more intuitive, more innovative and more elegant approach is ultimately needed to drive the adoption of model-based simulation and visualization in the larger community. Such an approach must rely on the computational power of a high performance computing foundation, along with powerful communication of large data sets via grid systems and embedded cyberinfrastructure, and visualization through next generation graphics and geographic information systems. The WATERS Network will thus need to emphasize both the development of advanced multi-disciplinary models, the incorporation of cyberinfrastructure services such as data and metadata stores, workflow systems, and services interfaces, and new interaction mechanisms for working with simulated environments.

5. Digital Libraries

Digital Libraries (DLs) abound (see for example

- the National Science Digital Library (NSDL, http://nsdl.org/)

- the Digital Library for Earth Science Education (DLESE, http://dlese.org/dds/index.jsp).

At their core, they are not very different from any other library, i.e. they store documents, albeit in digital form and not as paper, tape, container, or any other "touchable" medium. DLs can however store a wider variety of digitized information including interactive materials (e.g. training exercises) and digital data files (all sorts of formats) and house significantly more material than their physical counterparts. In some sense, DLs can be considered as storage for "piles of files", i.e. a general warehouse for all things digital that have been collected and need to be preserved in a storage facility, but the addition of interactive materials, the need for enhanced indexing given the anticipated volume of materials, the connection to knowledge management, the expectation of use both by humans and software, and the issues related to long-term curation of digital materials (e.g. migration of data to new physical media and supporting access to material created by obsolete programs) makes them significantly more useful and more difficult to design and manage.

There is a clear need for EOS to have a DL installed as part of their information system structure. What is less clear is whether there will be a clear distinction between the DL and the overall knowledge management and data/metadata handling capabilities. The DL must add support for the documents, presentations, and ancillary materials needed by the community, but it could also serve as collection point for already existing data that is being copied or newly introduced into the WATERS Network from other EOS-related projects (e.g. state/local agencies, NGOs, etc.). The DL could also be the basis for the sophisticated environments for analysis, fusion, visualization, and added-value processing. In this capacity the DL will:

- preserve data long term. The DLs are typically designed to be scalable, i.e. to grow over time to hold more and more data and to preserve it for future use.

- provide an extensive description framework (metadata) that allows users to query, find, access, and retrieve the data holdings for further processing. Metadata in this context is crucial given the diversity of the data holdings.

- be a first collection station of incoming and new data. Data can be stored and processed (for example QA/QC) before it is being released and/or passed on to other data holdings (e.g. special function data bases specifically designed to hold time series data).

In other words, DLs could also be the base for the EOS data storage and a central component for its information system architecture. As a result, it is important to realize that the aspect of interoperability with other EOS is crucial and must be addressed when designing and implementing a DL system.

Collaboration Services

The report of the NSF blue-ribbon panel on cyberinfrastructure (Atkins et al. 2003) cites the importance of collaboration services. Specifically, from the figure describing the components of cyberinfrastructure in that report (reproduced as Figure 1 of the main body of this report.), support for collaboration has an equal status with more traditional forms of cyberinfrastructure (CI), such as high performance computation, data, and visualization. Broadly speaking, collaboration services encompass tools for synchronous and asynchronous communication and collaboration. Familiar examples of asynchronous tools include email, while blogs and wikis are more modern instances. Familiar examples of synchronous tools include audio and video conferencing, while instant messaging (IM) is a more recent tool. By combining the synchronous/asynchronous dimensions with degree of collocation (same place vs. different place), it is possible to define a continuum of required capabilities that must be supported via collaboration services. Specifically, CI-mediated collaborations must succeed in each of the quadrants described in Table F1.

Each cell of Table F1 shows current collaboration approaches contrasted with potential future approaches. For example, the standard techniques used in most collaborations today include face-to-face interaction (synchronous and same place), email (asynchronous and different place), audio conferencing (synchronous and different place), and bulletin boards (asynchronous and same place). However, dramatic changes in the availability of network bandwidth have the potential to dramatically alter how people collaborate with both near and distant colleagues.

	Same place	Different place
Synchronous	Face-to-face	Audio conference
		HD videoconference
Asynchronous	Bulletin board	Email
	Multi-megapixel display	*Blogs/wikis*

For example, increased need for communication richness (i.e., immediacy of feedback, availability of multiple communication channels) may lead people to choose videoconferencing over audio conferencing when interacting at a distance (see F1). In the past year, for instance, it has become possible for the first time to host multi-way high definition (HD) videoconferences via the internet. Similarly, rather than tacking physical images and documents on a surface, collaborators may choose to use multi-megapixel displays, capable of rendering millions of pixels on wall-sized configurations of monitors (see F2). A closer connection between scientific applications and collaboration capabilities may also occur. For example, some scientific content is already being hosted within virtual collaborative worlds such as www.secondlife.com, and it is possible to imagine more science-centric collaboration ranging from sharing of scientific

Figure F1. HD video demonstration at iGrid 2005

workflows and annotation of data to much more game-like collaborative interactions around models and digital representations of physical environments.

A driving factor in development and adoption of collaboration services will be whether new tools and applications address critical needs within scientific and engineering collaborations. For example, a recent NSF report (Cummings and Kiesler, 2003), found that the greatest barrier to success within geographically-dispersed collaborations was the absence of strong coordination and communication tools. Dispersed collaborations underperformed relative to co-located collaborations largely because of lost effort due to increased coordination overhead. Examples of this increased overhead include more time spent to schedule meetings at a distance, inability to conveniently track and gauge progress on dependent projects, and barriers to trust formation as a function of reduced physical proximity.

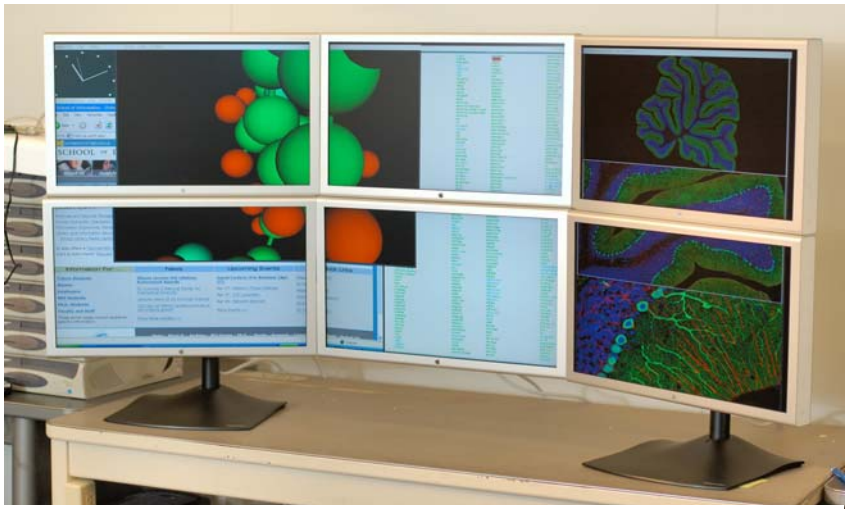


Figure F2. Multi-megapixel display

Collaboration services are something many researchers in the field are already familiar with and use with different degrees of intensity. Email communications allow more rapid discussion without in-person meetings. Web-sites with pre-prints or electronic copies of published journal articles allow faster access to existing lines of communication. Web-broadcasts allow participation in conferences and even giving presentations without physical presence. Online teaching environments allow rapid dissemination of materials and receipt of student feedback. Chat sessions provide near-real-time discussion of research or educational issues. To some extent, these services (all developed within the lifetime of most of the current crop of researchers) have revolutionized the way we do science. And yet, many of us now treat them as commonplace conveniences. At one level, collaboration services as envisioned in cyberinfrastructure are nothing more than an extension of these now commonplace tools. Group interface environments, rapid exchange of large datasets and the visualization results of scientists' interactions with those datasets are all seen as part of collaboration services. These are incremental benefits – allowing researchers to do more quickly and easily things that we know how to do and that are currently time-consuming and tedious (much like the auto spell checker eliminated the need to get up and walk across the room for the dictionary). But, the vision of CI includes a much deeper role for collaboration services. Collaboration services are envisioned as allowing "collective visualization and manipulation of data in real time" (NCAR 2003) which again emphasizes the interest in making collaboration an embedded part of scientific research.

As an example, consider the field of PCB research on the Hudson River. A number of research groups are currently working with the extensive field data available for this important site. New techniques to visualize and analyze these data are developed routinely and make their way through the peer review process and into the literature. Occasionally presentations at conferences alert researchers to new models or new methods for analysis that are being developed by other researchers. But, what collaboration is possible on this problem is slowed by the pace of publication, the complex nature of the modeling environments (developed separately by each researcher on his/her own systems), and the incremental and cyclic process by which research is currently conducted (run the simulation, look at the results with the graduate student, think about the next simulation to run, REPEAT). Transfer of information and new developments from one group to another is slow. CI envisions removing these barriers by allowing seamless integration of newly developed models into the "collaboratory" where multiple researchers at different institutions could run them, post simulations, and discuss in near-real time their planned "next steps." This would enable extensive utilization of a new tool without the long time-lag of transferring the code and explaining its inner workings to a new collaborator each time a team is being formed across institutions.

Another example less specific to this domain is the issue of access to primary data. When a journal article is published, the limitations of space and exigencies of publishing require encapsulating many experiments into summary figures and representative tables. Building on this research; however, often requires access to the complete set of primary data. Recently some journals have begun to allow authors to include "supplementary information" for web-only versions of articles. Thus, the primary journal article is supplemented by additional information accessible to the reader through the internet. This is an early form of what collaboration services are designed to provide. The ability to access primary raw data for re-analysis, expansion, evaluation, comparison with other data sets or incorporation into larger data mining investigations will be a significant contribution of cyberinfrastructure to building collaborative community networking.

Appendix G. List of Acronyms

AEESP – Association of Environmental Engineering and Science Professors

AmeriFlux = AmeriFlux network

BIRN = Biomedical Informatics Research Network

CUAHSI = Consortium of Universities for Advancement of Hydrologic Science

CISE – NSF Directorate for Computer and Information Science

CI = Cyberinfrastructure

CLEANER = Collaborative Large-scale Engineering Analysis Network for Environmental Research

CMCS

COTS = Commercial off-the-shelf

DAYMET = a model that generates daily surfaces of temperature, precipitation, humidity, and radiation over large regions of complex terrain

DOD =Department of Defense

DOE = Department of Energy

Earth Systems Grid

EON= Environmental Observatory Network

ESRI = GIS and mapping software company

GEON = Geosciences Network

Globus = Globus Alliance - a community of organizations and individuals developing fundamental technologies behind the "Grid,"

GriPhyN = Grid Physics Network

HCI = Human computer interaction

HIS = Hydrologic Information Systems

HPC = high performance computing

iVDGL = International Virtual Data Grid Laboratory

KDI = Knowledge and Distributed Intelligence

LOOKING

MODIS = Moderate-resolution Imaging Spectroradiometer

NASA = National Aeronautics and Space Administration

NAWQA = National Water-Quality Assessment Program

NCSA = National Center for Supercomputing Applications

NEES= Network for Earthquake Engineering Simulation

NEON = National Ecological Observatory Network

NIH = National Institutes of Health

NOAA = National Oceanic and Atmospheric Administration

NRC = National Research Council

NSFNET = National Science Foundation Network

NSF= U.S. National Science Foundation

NWIS = National Water Information System

NVO

OCI = Office of Cyberinfrastructure

OOI= Ocean Observatory Initiative

OptiPuter = named for its use of **Opt**ical networking, **I**nternet **P**rotocol, comp**uter** storage,

PACI= Partnership for Advanced Computation Infrastructure

QA/QC = Quality Assurance / Quality Control

ROADnet = Real-time Observatories, Applications, and Data management Network

SBE = NSF Directorate for Social, Behavioral, and Economic Sciences (NSF)

SDSC = San Diego Supercomputing Center

SEEK = Science Environment for Ecological Knowledge

STORET = USEPA repository for water quality short for STOrage and RETrieval

TeraGrid = Extensible Terascale Facility

UTC =

USDA = US Department of Agriculture

USGS = US Geological Survey

USEPA = US Environmental Protection Agency

vBNS = very high speed Backbone Network Services

W3C = World Wide Web Consortium

WATERS = WATer and Environmental Research Systems

Appendix H. WATERS Network Cyberinfrastructure Committee

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